Minqi's Solutions to the Exercises of Sheldon Axler (2014), Linear Algebra Done Right (3rd Edition)

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Contents

Preface ix

Chapter 1. $\mathbb{R}^n$ and $\mathbb{C}^n$ 1
1.1. Exercise 1.A.1: Multiplicative Inverse of Complex Numbers 1
1.2. Exercise 1.A.2: A Cubic Root of 1 1
1.3. Exercise 1.A.3: Square Roots of i 2
1.4. Exercise 1.A.4: Commutativity of Addition of Complex Numbers 2
1.5. Exercise 1.A.5: Associativity of Addition of Complex Numbers 2
1.6. Exercise 1.A.6: Associativity of Multiplication of Complex Numbers 2
1.7. Exercise 1.A.7: Existence and Uniqueness of Additive Inverse 3
1.8. Exercise 1.A.8: Existence and Uniqueness of Multiplicative Inverse 3
1.9. Exercise 1.A.9: The Distributive Property of Complex Numbers 3
1.10. Exercise 1.A.10: Vector Addition and Scalar Multiplication in $\mathbb{R}^4$ 4
1.11. Exercise 1.A.11: A Scalar Multiplication Example in $\mathbb{C}^3$ 4
1.12. Exercise 1.A.12: Associativity of Vector Addition in $\mathbb{F}^n$ 4
1.13. Exercise 1.A.13: Associativity of Scalar Multiplication in $\mathbb{F}^n$ 5
1.14. Exercise 1.A.14: Scalar Multiplication by 1 in $\mathbb{F}^n$ 5
1.15. Exercise 1.A.15: Left-distributive Property of Operations in $\mathbb{F}^n$ 5
1.16. Exercise 1.A.16: Right-distributive Property of Operations in $\mathbb{F}^n$ 5

Chapter 2. Definition of Vector Space 7
2.1. Exercise 1.B.1: $-(-v) = v$ in Vector Spaces 7
2.2. Exercise 1.B.2: $av = 0$ Implies $a = 0$ or $v = 0$ in Vector Spaces 7
2.3. Exercise 1.B.3: Solutions of a Vector Equation 7
2.4. Exercise 1.B.4: The Empty Set is NOT a Vector Space 8
2.5. Exercise 1.B.5: An Alternative Definition of Vector Space 8
2.6. Exercise 1.B.6: A Definition of $\mathbb{R} \cup \{\infty\} \cup \{-\infty\}$ 8

Chapter 3. Subspaces 11
3.1. Exercise 1.C.1: Examples and Counterexamples of Subspaces of $\mathbb{F}^3$ 11
3.2. Exercise 1.C.2: Supplying Proofs for Examples of Vector Spaces 12
3.3. Exercise 1.C.3: A Subspace of $\mathbb{R}^{(-4,4)}$ 13
3.4. Exercise 1.C.4: A Subspace of $\mathbb{R}^{[0,1]}$ 13
3.5. Exercise 1.C.5: $\mathbb{R}^2$ is NOT a Subspace of $\mathbb{C}^2$ 13
3.6. Exercise 1.C.6: Comparing Subspaces of $\mathbb{R}^3$ and $\mathbb{C}^3$ 13
3.7. Exercise 1.C.7: A Quasi-subspace Unclosed under Scalar Multiplication 14
3.8. Exercise 1.C.8: A Quasi-subspace Unclosed under Vector Addition 14
3.9. Exercise 1.C.9: Periodic Functions is NOT a Subspace of $\mathbb{R}^{\mathbb{R}}$ 14
3.10. Exercise 1.C.10: The Intersection of 2 Subspaces is a Subspace 15
3.11. Exercise 1.C.11: The Intersection of Subspaces is a Subspace 15

3.12. Exercise 1.C.12: When the Union of 2 Subspaces is a Subspace 15
3.13. Exercise 1.C.13: When the Union of 3 Subspaces is a Subspace 16
3.14. Exercise 1.C.14: An Example of Sums of Subspaces 16
3.15. Exercise 1.C.15: $U + U = U$ for Any Subspace U 17
3.16. Exercise 1.C.16: Commutativity of the Sum of Subspaces 17
3.17. Exercise 1.C.17: Associativity of the Sum of Subspaces 17
3.18. Exercise 1.C.18: The Identity of Sum of Subspaces 18
3.19. Exercise 1.C.19: There is NO Cancellation Law for Sums of Subspaces 18
3.20. Exercise 1.C.20: A Direct Sum of 2 Subspaces that Equals to $\mathbb{F}^4$ 18
3.21. Exercise 1.C.21: A Direct Sum of 2 Subspaces that Equals to $\mathbb{F}^5$ 19
3.22. Exercise 1.C.22: A Direct Sum of 4 Subspaces that Equals to $\mathbb{F}^5$ 19
3.23. Exercise 1.C.23: There is NO Cancellation Law for Direct Sums 20
3.24. Exercise 1.C.24: A Direct Sum Decomposition of $\mathbb{R}^{\mathbb{R}}$ 20

Chapter 4. Span and Linear Independence 23
4.1. Exercise 2.A.1: Alternative Span from an Existing Span 23
4.2. Exercise 2.A.2: Examples of Linearly Independent Lists 24
4.3. Exercise 2.A.3: a Linearly Dependent List 25
4.4. Exercise 2.A.4: Conditions to Linear Dependence 25
4.5. Exercise 2.A.5: Difference between $\mathbb{C}$ and $\mathbb{R}$ as Scalar Fields 25
4.6. Exercise 2.A.6: Constructing Another Independent List from a List 26
4.7. Exercise 2.A.7: Constructing Another Independent List from a List 27
4.8. Exercise 2.A.8: Linear Independent List Multiplied by Scalar 27
4.9. Exercise 2.A.9: Addition of Two Linearly Independent Lists 28
4.10. Exercise 2.A.10: Adding a Vector to a Linearly Independent List 28
4.11. Exercise 2.A.11: Concatenating a Vector to an Independent List 28
4.12. Exercise 2.A.12: Possible Linearly Independent Lists in $\mathcal{P}_4(\mathbb{F})$ 29
4.13. Exercise 2.A.13: Possible Spans of $\mathcal{P}_4(\mathbb{F})$ 29
4.14. Exercise 2.A.14: Another Definition of Infinite-Dimensional 29
4.15. Exercise 2.A.15: $\mathbb{F}^\infty$ is Infinite-Dimensional 29
4.16. Exercise 2.A.16: $\mathbb{R}^{[0,1]}$ is Infinite-Dimensional 30
4.17. Exercise 2.A.17: Polynomials that Vanish at the Same Point 30

Chapter 5. Bases 33
5.1. Exercise 2.B.1: Vector Spaces that Have Exactly One Basis 33
5.2. Exercise 2.B.2: Examples and Counterexamples of Bases 33
5.3. Exercise 2.B.3: Find and Extend a Basis for a Subspace of $\mathbb{R}^5$ 34
5.4. Exercise 2.B.4: Find and Extend a Basis for a Subspace of $\mathbb{C}^5$ 35
5.5. Exercise 2.B.5: A Basis of $\mathcal{P}_3(\mathbb{F})$ without Degree-2 Polynomials 36
5.6. Exercise 2.B.6: Constructing Another Basis from a Basis 38
5.7. Exercise 2.B.7: Constructing a Counterexample 39
5.8. Exercise 2.B.8: Basis Obtained from a Direct Sum 40

Chapter 6. Dimension 41
6.1. Exercise 2.C.1: A Space with the Same Dimension of its Subspace 41
6.2. Exercise 2.C.2: Subspaces of $\mathbb{R}^2$ 41
6.3. Exercise 2.C.3: Subspaces of $\mathbb{R}^3$ 41
6.4. Exercise 2.C.4: Find and Extend a Basis for a Subspace of $\mathcal{P}_4(\mathbb{F})$ 42
6.5. Exercise 2.C.9: $\dim \operatorname{span}(v_1 + w, \ldots, v_m + w) \geqslant m - 1$ 43

Chapter 7. The Vector Space of Linear Maps 45
7.1. Exercise 3.A.1: Conditions for a Mapping to be Linear 45
7.2. Exercise 3.A.5: $\mathcal{L}(V, W)$ is a vector space 46
7.3. Exercise 3.A.6: Algebraic Properties of Products of Linear Maps 46
7.4. Exercise 3.A.7: Linear Map on 1D Space is Scalar Multiplication 47
7.5. Exercise 3.A.8: Homogeneity Alone does NOT Imply Linearity 47
7.6. Exercise 3.A.9: Additivity Alone does NOT Imply Linearity 48

Chapter 8. Null Spaces and Ranges 49
8.1. Exercise 3.B.1: A Linear Map T with $\dim \operatorname{null} T = 3, \dim \operatorname{range} T = 2$ 49
8.2. Exercise 3.B.2: range $S \subset$ null T Implies $(ST)^2 = 0$ 49
8.3. Exercise 3.B.3: Spanning = Surjective; Independent = Injective 49
8.4. Exercise 3.B.4: $\{T \in \mathcal{L}(\mathbb{R}^5, \mathbb{R}^4) : \dim \operatorname{null} T > 2\}$ is NOT a Subspace 50
8.5. Exercise 3.B.5: A Linear Map T such that range $T =$ null T 50

Chapter 9. Matrices 53
9.1. Exercise 3.C.1: $\mathcal{M}(T)$ Has At Least $\dim \operatorname{range} T$ Nonzero Entries 53
9.2. Exercise 3.C.2: Find Bases from the Matrix of $D \in \mathcal{L}(\mathcal{P}_3(\mathbb{R}), \mathcal{P}_2(\mathbb{R}))$ 53
9.3. Exercise 3.C.3: Find Bases for V, W from $\mathcal{M}(T)$ where $T \in \mathcal{L}(V, W)$ 54
9.4. Exercise 3.C.4: Find Bases for W from $\mathcal{M}(T)$ where $T \in \mathcal{L}(V, W)$ 54
9.5. Exercise 3.C.5: Find Bases for V from $\mathcal{M}(T)$ where $T \in \mathcal{L}(V, W)$ 54

Chapter 10. Invertibility and Isomorphic Vector Spaces 57
10.1. Exercise 3.D.1: $(ST)^{-1} = T^{-1}S^{-1}$ 57
10.2. Exercise 3.D.2: Noninvertible Operators on V is NOT a Subspace 57
10.3. Exercise 3.D.3: Injective Subspace Mappings and Invertible Operators 58
10.4. Exercise 3.D.4: null $T_1 =$ null T_2 if and only if $T_1 = ST_2$ 58
10.5. Exercise 3.D.5: range $T_1 =$ range T_2 if and only if $T_1 = T_2S$ 59

Chapter 11. Products and Quotients of Vector Spaces 61
11.1. Exercise 3.E.1: T is a Linear Map iff graph of T is a Subspace of $V \times W$ 61
11.2. Exercise 3.E.2: V_j is Finite-dimensional if $V_1 \times \cdots \times V_m$ is Finite-dimensional 61
11.3. Exercise 3.E.3: Example of $U_1 \times U_2$ Isomorphic to Non-Direct-Sum $U_1 + U_2$ 62
11.4. Exercise 3.E.4: $\mathcal{L}(V_1 \times \cdots \times V_m, W)$ and $\mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W)$ are Isomorphic 62
11.5. Exercise 3.E.5: $\mathcal{L}(V, W_1 \times \cdots \times W_m)$ and $\mathcal{L}(V, W_1) \times \cdots \times \mathcal{L}(V, W_m)$ are Isomorphic 63

Chapter 12. Duality 65
12.1. Exercise 3.F.1: Linear Functionals are either Surjective or Zero 65
12.2. Exercise 3.F.2: Examples of Linear Functionals on $\mathbb{R}^{[0,1]}$ 65

Bibliography 67

Preface

Remarkable exercises are:

(1) Exercise 1.C.9: Periodic Functions is NOT a Subspace of $\mathbb{R}^{\mathbb{R}}$.
(2) Exercise 1.C.13: When the Union of 3 Subspaces is a Subspace.
(3) Exercise 2.B.7: Constructing a Counterexample.
(4) Exercise 2.C.9: $\dim \operatorname{span}(v_1 + w, \ldots, v_m + w) \geqslant m - 1$.
(5) Exercise 3.C.5: Find Bases for V from $\mathcal{M}(T)$ where $T \in \mathcal{L}(V, W)$.
(6) Exercise 3.D.4: null $T_1 =$ null T_2 if and only if $T_1 = ST_2$.
(7) Exercise 3.D.5: range $T_1 =$ range T_2 if and only if $T_1 = T_2S$.

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CHAPTER 1

$\mathbb{R}^n$ and $\mathbb{C}^n$

1.1. Exercise 1.A.1: Multiplicative Inverse of Complex Numbers

Suppose a and b are real numbers, not both 0. Find real numbers c and d such that

$$1/(a+bi) = c + di$$

Answer. Since at least one of the real numbers a, b is different from 0, we have $a^2 + b^2 \neq 0$. We can then let

$$c = \frac{a}{a^2+b^2},$$

$$d = \frac{-b}{a^2+b^2}.$$

Then

$$\begin{aligned}
&(a+bi)\left(\frac{a}{a^2+b^2} - \frac{b}{a^2+b^2}i\right) \\
&= \left(a\frac{a}{a^2+b^2} - b\left(-\frac{b}{a^2+b^2}\right)\right) + \left(a\left(-\frac{b}{a^2+b^2}\right) + b\frac{a}{a^2+b^2}\right)i \\
&= 1 + 0i \\
&= 1.
\end{aligned}$$

□

1.2. Exercise 1.A.2: A Cubic Root of 1

Show that

$$\frac{-1+\sqrt{3}i}{2}$$

is a cube root of 1 (meaning that its cube equals 1).

Answer.

$$\begin{aligned}
&\frac{-1+\sqrt{3}i}{2} \cdot \frac{-1+\sqrt{3}i}{2} \cdot \frac{-1+\sqrt{3}i}{2} \\
&= \frac{-2-2\sqrt{3}i}{4} \cdot \frac{-1+\sqrt{3}i}{2} \\
&= \frac{(-2)(-1) - (-2\sqrt{3})(\sqrt{3}) + (-2\sqrt{3} + (-2\sqrt{3})(-1))i}{8} \\
&= 1
\end{aligned}$$

□

1.3. Exercise 1.A.3: Square Roots of i

Find two distinct square roots of i.

ANSWER. Note that

$$i = e^{i\frac{\pi}{2}}.$$

Thus the following two numbers are two distinct square roots of i:

$$\pm e^{i\frac{\pi}{4}} = \pm(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i).$$

□

1.4. Exercise 1.A.4: Commutativity of Addition of Complex Numbers

Show that $\alpha + \beta = \beta + \alpha$ for all $\alpha, \beta \in \mathbb{C}$.

PROOF. Suppose $\alpha = a + bi$ and $\beta = c + di$, where $a, b, c, d \in \mathbb{R}$. Then the definition of addition of complex numbers shows that

$$\alpha + \beta = (a + c) + (b + d)i$$

and

$$\beta + \alpha = (c + a) + (d + b)i.$$

The equations above and the commutativity of addition of real numbers show that $\alpha + \beta = \beta + \alpha$. □

1.5. Exercise 1.A.5: Associativity of Addition of Complex Numbers

Show that $(\alpha + \beta) + \lambda = \alpha + (\beta + \lambda)$ for all $\alpha, \beta, \lambda \in \mathbb{C}$.

PROOF. Suppose $\alpha = a + bi$, $\beta = c + di$ and $\lambda = e + fi$, where $a, b, c, d, e, f \in \mathbb{R}$. Then the definition of addition of complex numbers shows that

$$\begin{aligned}(\alpha + \beta) + \lambda &= ((a + c) + (b + d)i) + e + fi \\ &= ((a + c) + e) + ((b + d) + f)i\end{aligned}$$

and

$$\begin{aligned}\alpha + (\beta + \lambda) &= a + bi + ((c + e) + (d + f)i) \\ &= (a + (c + e)) + (b + (d + f))i.\end{aligned}$$

The equations above and the associativity of addition of real numbers show that $(\alpha + \beta) + \lambda = \alpha + (\beta + \lambda)$. □

1.6. Exercise 1.A.6: Associativity of Multiplication of Complex Numbers

Show that $(\alpha\beta)\lambda = \alpha(\beta\lambda)$ for all $\alpha, \beta, \lambda \in \mathbb{C}$.

PROOF. Suppose $\alpha = a + bi$, $\beta = c + di$ and $\lambda = e + fi$, where $a, b, c, d, e, f \in \mathbb{R}$. Then the definition of multiplication of complex numbers shows that

$$\begin{aligned}(\alpha\beta)\lambda &= ((ac - bd) + (ad + bc)i)(e + fi) \\ &= ((ac - bd)e - (ad + bc)f) + ((ac - bd)f + (ad + bc)e)i \\ &= (ace - bde - adf - bcf) + (acf - bdf + ade + bce)i\end{aligned}$$

and

$$\begin{aligned}\alpha(\beta\lambda) &= (a+bi)((ce-df)+(cf+de)i)\\ &= (a(ce-df)-b(cf+de))+(a(cf+de)+b(ce-df))i\\ &= (ace-adf-bcf-bde)+(acf+ade+bce-bdf)i.\end{aligned}$$

The equations above and the commutativity of addition of real numbers show that $(\alpha\beta)\lambda = \alpha(\beta\lambda)$. □

1.7. Exercise 1.A.7: Existence and Uniqueness of Additive Inverse

Show that for every $\alpha \in \mathbb{C}$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha + \beta = 0$.

Proof. Suppose $\alpha = a + bi$ where $a, b \in \mathbb{R}$, then $\beta = -a - bi$ satisfies $\alpha + \beta = 0$.
Suppose that there exists another $\beta' \in \mathbb{C}$ such that $\alpha + \beta' = 0$. Then

$$\begin{aligned}\beta &= \beta + (\alpha + \beta')\\ &= (\beta + \alpha) + \beta'\\ &= \beta'.\end{aligned}$$

The equations above show that $\beta' = \beta$. □

1.8. Exercise 1.A.8: Existence and Uniqueness of Multiplicative Inverse

Show that for every $\alpha \in \mathbb{C}$ with $\alpha \neq 0$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha\beta = 1$.

Proof. The existence of β has been shown in Exercise 1.A.1.
Suppose that there exists another $\beta' \in \mathbb{C}$ such that and $\alpha\beta' = 1$. Then

$$\begin{aligned}\beta &= \beta(\alpha\beta')\\ &= (\beta\alpha)\beta'\\ &= \beta'.\end{aligned}$$

The equations above show that $\beta' = \beta$. □

1.9. Exercise 1.A.9: The Distributive Property of Complex Numbers

Show that $\lambda(\alpha + \beta) = \lambda\alpha + \lambda\beta$ for all $\lambda, \alpha, \beta \in \mathbb{C}$.

Proof. Suppose $\alpha = a + bi$, $\beta = c + di$ and $\lambda = e + fi$, where $a, b, c, d, e, f \in \mathbb{R}$. Then the definition of multiplication and addition of complex numbers shows that

$$\begin{aligned}\lambda(\alpha+\beta) &= (e+fi)((a+c)+(b+d)i)\\ &= (e(a+c)-f(b+d))+(e(b+d)+f(a+c))i\\ &= (ea+ec-fb-fd)+(eb+ed+fa+fc)i\end{aligned}$$

and

$$\begin{aligned}\lambda\alpha+\lambda\beta &= (e+fi)(a+bi)+(e+fi)(c+di)\\ &= ((ea-fb)+(eb+fa)i)+((ec-fd)+(ed+fc)i)\\ &= (ea-fb+ec-fd)+(eb+fa+ed+fc)i.\end{aligned}$$

The equations above and the commutativity of addition of real numbers show that $\lambda(\alpha+\beta) = \lambda\alpha + \lambda\beta$. □

1.10. Exercise 1.A.10: Vector Addition and Scalar Multiplication in $\mathbb{R}^4$

Find $x \in \mathbb{R}^4$ such that

$$(4, -3, 1, 7) + 2x = (5, 9, -6, 8).$$

Answer.

$$x = \left(\frac{1}{2}, 6, -\frac{7}{2}, \frac{1}{2}\right).$$

□

1.11. Exercise 1.A.11: A Scalar Multiplication Example in $\mathbb{C}^3$

Explain why there does not exist $\lambda \in \mathbb{C}$ such that

$$\lambda(2 - 3i, 5 + 4i, -6 + 7i) = (12 - 5i, 7 + 22i, -32 - 9i).$$

Proof. If there exists such λ then the definition of scalar multiplication in $\mathbb{C}^3$ shows that,

$$\lambda(2 - 3i) = 12 - 5i.$$

Thus

$$\lambda = \frac{12 - 5i}{2 - 3i} = 3 + 2i.$$

But

$$\begin{aligned}\lambda(2 - 3i, 5 + 4i, -6 + 7i) &= (3 + 2i)(2 - 3i, 5 + 4i, -6 + 7i)\\ &= (12 - 5i, 7 + 22i, -32 + 9i)\\ &\neq (12 - 5i, 7 + 22i, -32 - 9i).\end{aligned}$$

Therefore λ does not exist. □

1.12. Exercise 1.A.12: Associativity of Vector Addition in $\mathbb{F}^n$

Show that $(x + y) + z = x + (y + z)$ for all $x, y, z \in \mathbb{F}^n$.

Proof. Suppose $x = (x_1, \ldots, x_n)$, $y = (y_1, \ldots, y_n)$ and $z = (z_1, \ldots, z_n)$ where $x_1, \ldots, x_n, y_1, \ldots, y_n, z_1, \ldots, z_n$ $\mathbb{F}$. Then the definition of addition in $\mathbb{F}^n$ shows that

$$\begin{aligned}(x + y) + z &= (x_1 + y_1, \ldots, x_n + y_n) + (z_1, \ldots, z_n)\\ &= ((x_1 + y_1) + z_1, \ldots, (x_n + y_n) + z_n)\end{aligned}$$

and

$$\begin{aligned}x + (y + z) &= (x_1, \ldots, x_n) + (y_1 + z_1, \ldots, y_n + z_n)\\ &= (x_1 + (y_1 + z_1), \ldots, x_n + (y_n + z_n))\end{aligned}$$

The equations above and the associativity of addition in $\mathbb{F}$ show that $(x + y) + z = x + (y + z)$. □

1.13. Exercise 1.A.13: Associativity of Scalar Multiplication in $\mathbb{F}^n$

Show that $(ab)x = a(bx)$ for all $x \in \mathbb{F}^n$ and all $a, b \in \mathbb{F}$.

PROOF. Suppose $x = (x_1, \ldots, x_n)$ where $x_1, \ldots, x_n \in \mathbb{F}$. Then the definition of scalar multiplication in $\mathbb{F}^n$ shows that

$$(ab)x = ((ab)x_1, \ldots, (ab)x_n)$$

and

$$\begin{aligned} a(bx) &= a(bx_1, \ldots, bx_n) \\ &= (a(bx_1), \ldots, a(x_n)) \end{aligned}$$

The equations above and the associativity of multiplication in $\mathbb{F}$ show that $(ab)x = a(bx)$. □

1.14. Exercise 1.A.14: Scalar Multiplication by 1 in $\mathbb{F}^n$

Show that $1x = x$ for all $x \in \mathbb{F}^n$.

PROOF. Suppose $x = (x_1, \ldots, x_n)$ where $x_1, \ldots, x_n \in \mathbb{F}$. Then the definition of scalar multiplication in $\mathbb{F}^n$ and the fact that 1 is the multiplicative identity of $\mathbb{F}$ shows that

$$\begin{aligned} 1x &= (1x_1, \ldots, 1x_n) \\ &= (x_1, \ldots, x_n) \\ &= x. \end{aligned}$$

□

1.15. Exercise 1.A.15: Left-distributive Property of Operations in $\mathbb{F}^n$

Show that $\lambda(x + y) = \lambda x + \lambda y$ for all $\lambda \in \mathbb{F}$ and $x, y \in \mathbb{F}^n$.

PROOF. Suppose $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$ where $x_1, \ldots, x_n \in \mathbb{F}$ and $y_1, \ldots, y_n \in \mathbb{F}$. Then the definition of addition and scalar multiplication in $\mathbb{F}^n$ shows that

$$\begin{aligned} \lambda(x + y) &= \lambda(x_1 + y_1, \ldots, x_n + y_n) \\ &= (\lambda(x_1 + y_1), \ldots, \lambda(x_n + y_n)) \end{aligned}$$

and

$$\begin{aligned} \lambda x + \lambda y &= (\lambda x_1, \ldots, \lambda x_n) + (\lambda y_1, \ldots, \lambda y_n) \\ &= (\lambda x_1 + \lambda y_1, \ldots, \lambda x_n + \lambda y_n) \end{aligned}$$

The equations above and the distributive property of $\mathbb{F}$ show that $\lambda(x + y) = \lambda x + \lambda y$. □

1.16. Exercise 1.A.16: Right-distributive Property of Operations in $\mathbb{F}^n$

Show that $(a + b)x = ax + bx$ for all $a, b \in \mathbb{F}$ and all $x \in \mathbb{F}^n$.

PROOF. Suppose $x = (x_1, \ldots, x_n)$ where $x_1, \ldots, x_n \in \mathbb{F}$. Then the definition of scalar multiplication and addition in $\mathbb{F}^n$ shows that

$$(a + b)x = ((a + b)x_1, \ldots, (a + b)x_n)$$

and

$$\begin{aligned} ax + bx &= (ax_1, \ldots, ax_n) + (bx_1, \ldots, bx_n) \\ &= (ax_1 + bx_1, \ldots, ax_n + bx_n). \end{aligned}$$

The equations above and the distributive property of $\mathbb{F}$ show that $(a + b)x = ax + bx$. □

CHAPTER 2

Definition of Vector Space

2.1. Exercise 1.B.1: $-(-v) = v$ in Vector Spaces

Prove that $-(-v) = v$ for every $v \in V$.

PROOF.

$$\begin{aligned} -(-v) &= -(-v) + 0 \\ &= -(-v) + ((-v) + v) \\ &= (-(-v) + (-v)) + v \\ &= 0 + v \\ &= v \end{aligned}$$

□

2.2. Exercise 1.B.2: $av = 0$ Implies $a = 0$ or $v = 0$ in Vector Spaces

Suppose $a \in \mathbb{F}, v \in V$, and $av = 0$. Prove that $a = 0$ or $v = 0$.

PROOF. For the purpose of arriving at a contradiction, assume that both $a \neq 0$ and $v \neq 0$. Then

$$v = 1v = (\frac{1}{a}a)v = \frac{1}{a}(av) = 0$$

which contradicts with $v \neq 0$.

Therefore $a = 0$ or $v = 0$. □

2.3. Exercise 1.B.3: Solutions of a Vector Equation

Suppose $v, w \in V$. Explain why there exists a unique $x \in V$ such that $v + 3x = w$.

PROOF. We claim that the following $x \in V$ satisfies $v + 3x = w$:

$$x = \frac{1}{3}(w - v).$$

We can verify it by showing that,

$$\begin{aligned} v + 3x &= v + 3 \times \frac{1}{3}(w - v) \\ &= v + w - v \\ &= w. \end{aligned}$$

Suppose that there exists another $x' \in V$ that also satisfies $v + 3x' = w$, then

$$\begin{aligned} 0 &= w - w \\ &= (v + 3x) - (v + 3x') \\ &= v + 3x - v - 3x' \\ &= 3x - 3x'. \end{aligned}$$

Thus

$$\begin{aligned} x' &= x' + \frac{1}{3} \times 0 \\ &= x' + \frac{1}{3}(3x - 3x') \\ &= x' + x - x' \\ &= x. \end{aligned}$$

□

2.4. Exercise 1.B.4: The Empty Set is NOT a Vector Space

The empty set is not a vector space. The empty set fails to satisfy only one of the requirements listed in 1.19. Which one?

Answer. The empty set fails to satisfy the "Additive Identity" requirement. Because there does not exist an element $0 \in \varnothing$. □

2.5. Exercise 1.B.5: An Alternative Definition of Vector Space

Show that in the definition of a vector space (1.19), the additive inverse condition can be replaced with the condition that

$$0v = 0, \forall v \in V.$$

Here the 0 on the left side is the number 0, and the 0 on the right side is the additive identity of V.

Proof. We will show that the new condition is equal to the original condition, which states that for every $v \in V$, there exists $w \in V$ such that $v + w = 0$. That the old condition implies the new condition has been shown in Theorem 1.29 "The number 0 times a vector". Therefore it suffices to show that the new condition implies the old condition.

Suppose that the new condition holds.

Pick $v \in V$, and let $w = (-1)v$. It follows that

$$\begin{aligned} v + w &= 1v + (-1)v \\ &= (1 - 1)v \\ &= 0v \\ &= 0. \end{aligned}$$

Therefore the old condition holds. □

2.6. Exercise 1.B.6: A Definition of $\mathbb{R} \cup \{\infty\} \cup \{-\infty\}$

Let ∞ and $-\infty$ denote two distinct objects, neither of which is in $\mathbb{R}$. Define an addition and scalar multiplication on $\mathbb{R} \cup \{\infty\} \cup \{-\infty\}$ as you could guess from the notation. Specifically, the sum and product of two real numbers is as usual, and for $t \in \mathbb{R}$ define

$$t\infty = \begin{cases} -\infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ \infty & \text{if } t > 0, \end{cases} \quad t(-\infty) = \begin{cases} \infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ -\infty & \text{if } t > 0, \end{cases}$$

$$t + \infty = \infty + t = \infty, \quad t + (-\infty) = (-\infty) + t = (-\infty),$$

$$\infty + \infty = \infty, \quad (-\infty) + (-\infty) = (-\infty), \quad \infty + (-\infty) = 0.$$

Is $\mathbb{R} \cup \{\infty\} \cup \{-\infty\}$ a vector space over $\mathbb{R}$? Explain.

Answer. No. Because the associativity of vector addition fails:

$$(1024 + \infty) - \infty = \infty - \infty = 0$$

but

$$1024 + (\infty - \infty) = 1024 + 0 = 1024.$$

$\square$

CHAPTER 3

Subspaces

3.1. Exercise 1.C.1: Examples and Counterexamples of Subspaces of $\mathbb{F}^3$

For each of the following subsets of $\mathbb{F}^3$, determine whether it is a subspace of $\mathbb{F}^3$:

(a) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}$;
(b) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 4\}$;
(c) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 x_2 x_3 = 0\}$;
(d) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 = 5x_3\}$.

Answer. (a) Denote $A = \{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}$. We claim that A is a subspace of $\mathbb{F}^3$ with the following verifications.

(i) $(0, 0, 0) \in A$ because $0 + 2 \cdot 0 + 3 \cdot 0 = 0$.

(ii) Pick $u, w \in A$, where $u = (u_1, u_2, u_3), w = (w_1, w_2, w_3)$. It follows that $u + w = (u_1 + w_1, u_2 + w_2, u_3 + w_3)$. We can show that $u + w \in A$ by the following equations:

$$\begin{aligned}&(u_1 + w_1) + 2(u_2 + w_2) + 3(u_3 + w_3)\\&= (u_1 + 2u_2 + 3u_3) + (w_1 + 2w_2 + 3w_3)\\&= 0 + 0 = 0.\end{aligned}$$

(iii) Pick $a \in \mathbb{F}$ and $u \in A$, where $u = (u_1, u_2, u_3)$. It follows that $au = (au_1, au_2, au_3)$. We can show that $au \in A$ by the following equations:

$$\begin{aligned}(au_1) + 2(au_2) + 3(au_3) &= a(u_1 + 2u_2 + 3u_3)\\&= a0 = 0.\end{aligned}$$

(b) Denote $B = \{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 4\}$. B is not a subspace of $\mathbb{F}^3$, because vector $(0, 0, 0) \notin B$ (as $0 + 2 \cdot 0 + 3 \cdot 0 \neq 4$).

(c) Denote $C = \{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 x_2 x_3 = 0\}$. C is not a subspace of $\mathbb{F}^3$ because by setting $u = (0, 1, 1) \in C$ and $w = (1, 1, 0) \in C$, we find out that $u + w = (1, 2, 1) \notin C$.

(d) Denote $D = \{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 = 5x_3\}$. We claim that D is a subspace of $\mathbb{F}^3$ with the following verifications.

(i) $(0, 0, 0) \in A$ because $0 = 5 * 0$.

(ii) Pick $u, w \in A$, where $u = (u_1, u_2, u_3), w = (w_1, w_2, w_3)$. It follows that $u + w = (u_1 + w_1, u_2 + w_2, u_3 + w_3)$. We can show that $u + w \in A$ by the following equations:

$$\begin{aligned}u_1 + w_1 &= 5u_3 + 5w_3\\&= 5(u_1 + w_3)\end{aligned}$$

(iii) Pick $a \in \mathbb{F}$ and $u \in A$, where $u = (u_1, u_2, u_3)$. It follows that $au = (au_1, au_2, au_3)$. We can show that $au \in A$ by the following equations:

$$\begin{aligned}au_1 &= a(5u_3)\\&= 5(au_3).\end{aligned}$$

□

3.2. Exercise 1.C.2: Supplying Proofs for Examples of Vector Spaces

Verify all the assertions in Example 1.35.

(a) If $b \in \mathbb{F}$, then
$$\{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 : x_3 = 5x_4 + b\}$$
is a subspace of $\mathbb{F}^4$ if and only if $b = 0$.

(b) The set of continuous real-values functions on the interval $[0, 1]$ is a subspace of $\mathbb{R}^{[0,1]}$.

(c) The set of differentiable real-valued functions on $\mathbb{R}$ is a subspace of $\mathbb{R}^{\mathbb{R}}$

(d) The set of differentiable real-valued functions f on the interval $(0, 3)$ such that $f'(2) = b$ is a subspace of $\mathbb{R}^{(0,3)}$ if and only if $b = 0$.

(e) The set of all sequences of complex numbers with limit 0 is a subspace of $\mathbb{C}^\infty$.

Answer. (a) Denote $A = \{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 : x_3 = 5x_4 + b\}$.

If $b = 0$, then A is a subspace by a verification similar to Exercise 1.C.2 (d).

On the other hand, if A is a subspace, then $0 \in A$. It follows that
$$0 = 5 * 0 + b = b.$$

(b) Denote $B = \{f \in \mathbb{R}^{[0,1]} : f \text{ is continuous}\}$.

We can verify that $0 \in B$ because the constant function $f = 0$ is continuous.

Suppose that $u, w \in B$, we can verify that $(u + w) \in B$ because the addition of two complex continuous functions is continuous by Theorem 4.9 of [**Rud76**].

Suppose that $a \in \mathbb{F}$ and $u \in B$. Note that the constant function $f = a$ is continuous. We can verify that $au \in B$ because the multiplication of two continuous complex functions is continuous by Theorem 4.9 of [**Rud76**].

(c) Denote $C = \{f \in \mathbb{R}^{\mathbb{R}} : f \text{ is differentiable}\}$.

We can verify that $0 \in C$ because the constant function $f = 0$ is differentiable.

Suppose that $u, w \in C$, we can verify that $(u + w) \in C$ because the addition of two differentiable real functions is differentiable by Theorem 5.3 of [**Rud76**].

Suppose that $a \in \mathbb{R}$ and $u \in C$. Note that the constant function $a(x) = a$ is differentiable. We can verify that $au \in C$ because the multiplication of two differentiable real functions is differentiable by Theorem 5.3 of [**Rud76**].

(d) Denote $D = \{f \in \mathbb{R}^{(0,3)} : f \text{ is differentiable and } f'(2) = b\}$.

Suppose that $b = 0$. We claim that D is a subspace of $\mathbb{R}^{(0,3)}$ with the following verifications.

(i) $0 \in D$ because the constant function $0(x) = 0$ is differentiable and $0'(2) = 0$.

(ii) Suppose that $u, w \in D$. $(u + w)$ is also differentiable by Theorem 5.3 of [**Rud76**]. Also by Theorem 5.3 of [**Rud76**], $(u + v)'(2) = u'(2) + v'(2) = 0 + 0 = 0$. Thus $(u + w) \in D$.

(iii) Suppose that $a \in \mathbb{R}$ and $u \in D$. Note that the constant function $a(x) = a$ is differentiable, so au is differentiable by Theorem 5.3 of [**Rud76**]. Also by Theorem 5.3 of [**Rud76**], $(au)'(2) = a'(2)u(2) + a(2)u'(2) = 0 \times u(2) + a \times 0 = 0$. Thus $au \in D$.

On the other hand, suppose that D is a subspace. Then it implies that $0 \in D$ where $0(x) = 0$, and $0'(x) = 0$ implies that $0'(2) = b = 0$.

(e) Denote $E = \{z_n \in \mathbb{C}^\infty : \lim_{n\to\infty} z_n = 0\}$.

We can verify that $0_n \in E$ where $0_n = (0, 0, 0, \dots)$ because
$$\lim_{n\to\infty} 0_n = 0.$$

Suppose that $u_n, w_n \in E$, we can verify that $(u_n + w_n) \in C$ because $\lim_{n\to\infty}(u_n + w_n) = \lim_{n\to\infty} u_n + \lim_{n\to\infty} w_n = 0 + 0 = 0$ by Theorem 3.3 of [**Rud76**].

Suppose that $a \in \mathbb{F}$ and $u_n \in E$, we can verify that $au_n \in E$ because $\lim_{n\to\infty} au_n = a \lim_{n\to\infty} u_n = a0 = 0$ by Theorem 3.3 of [**Rud76**].

□

3.3. Exercise 1.C.3: A Subspace of $\mathbb{R}^{(-4,4)}$

Show that the set of differentiable real-valued functions f on the interval $(-4, 4)$ such that $f'(-1) = 3f(2)$ is a subspace of $\mathbb{R}^{(-4,4)}$.

Proof. Denote $X = \{f \in \mathbb{R}^{(-4,4)} : f \text{ is differentiable and } f'(-1) = 3f(2)\}$

(1) We claim that $0 \in X$ where $0(x) = 0$. Note that 0 is differentiable and $0'(-1) = 0, 0(2) = 0$, thus $0'(-1) = 3 \times 0(2)$.
(2) Pick $f, g \in X$. That $f + g$ is differentiable follows from Theorem 5.3 of [**Rud76**]. Also by Theorem 5.3 of [**Rud76**], $(f + g)'(-1) = f'(-1) + g'(-1) = 3f(2) + 3g(2) = 3(f(2) + g(2)) = 3((f + g)(2))$. Thus $f + g \in X$.
(3) Pick $f \in X$ and $a \in \mathbb{F}$. That af is differentiable follows from Theorem 5.3 of [**Rud76**] by considering a as a constant function $a(x) = a$. Also by Theorem 5.3 of [**Rud76**], $(af)'(-1) = a(f'(-1)) = a(3f(2)) = 3(af(2)) = 3((af)(2))$. Thus $af \in X$.

□

3.4. Exercise 1.C.4: A Subspace of $\mathbb{R}^{[0,1]}$

Suppose $b \in \mathbb{R}$. Show that the set of continuous real-valued functions f on the interval $[0, 1]$ such that $\int_0^1 f = b$ is a subspace of $\mathbb{R}^{[0,1]}$ if and only if $b = 0$.

Proof. Denote $X = \{f \in \mathbb{R}^{[0,1]} :\text{f is continuous and } \int_0^1 f = b\}$.
Suppose that $b = 0$. We now verify that X is a subspace.

(1) We can verify that $0 \in X$ because $0(x) = 0$ is continuous and $\int_0^1 0(x)dx = 0$.
(2) Pick $f, g \in X$. It follows that $f + g$ is continuous by Theorem 4.9 of [**Rud76**] and Riemann-integrable by Theorem 6.8 of [**Rud76**]. Further by Theorem 6.12 of [**Rud76**], we have $\int_0^1 (f(x) + g(x))dx = \int_0^1 f(x)dx + \int_0^1 g(x) = 0 + 0 = 0$.
(3) Pick $a \in \mathbb{F}, f \in X$. By Theorem 4.9 of [**Rud76**], af is continuous. Thus af is Riemann-integrable by Theorem 6.8 of [**Rud76**]. Further by Theorem 6.12 of [**Rud76**], we have $\int_0^1 (af(x))dx = a \int_0^1 f(x)dx = a \times 0 = 0$.

On the other hand, suppose that X is a subspace. Then it follows that $0 \in X$, which implies that $\int_0^1 0(x)dx = 0 = b$. □

3.5. Exercise 1.C.5: $\mathbb{R}^2$ is NOT a Subspace of $\mathbb{C}^2$

Is $\mathbb{R}^2$ a subspace of the complex vector space $\mathbb{C}^2$?

Answer. No. Because multiplying the vector $(1, 1) \in \mathbb{R}^2$ by scalar i gives (i, i) which is not in $\mathbb{R}^2$. □

3.6. Exercise 1.C.6: Comparing Subspaces of $\mathbb{R}^3$ and $\mathbb{C}^3$

(i) Is $\{(a, b, c) \in \mathbb{R}^3 : a^3 = b^3\}$ a subspace of $\mathbb{R}^3$?
(ii) Is $\{(a, b, c) \in \mathbb{C}^3 : a^3 = b^3\}$ a subspace of $\mathbb{C}^3$?

Answer. (i) Yes. Let's observe the real function $f(x) = x^3$. $f(x)$ is a 1-1 correspondence between $\mathbb{R}$ and $\mathbb{R}$. Therefore the condition $a^3 = b^3$ is equivalent to $a = b$.

(ii) No. Denote $X = \{(a,b,c) \in \mathbb{C}^3 : a^3 = b^3\}$. Set $u = (2, 2e^{\frac{2\pi}{3}i}, 0)$ and $v = (e^{\frac{2\pi}{3}i}, 1, 0)$, then $u+v = (2+e^{\frac{2\pi}{3}i}, 2e^{\frac{2\pi}{3}i}+1, 0)$. Note that $u \in X$ and $v \in X$ but $u+v \notin X$ because,

$$\left(2 + e^{\frac{2\pi}{3}i}\right)^3 = 3\sqrt{3}i,$$

$$\left(2e^{\frac{2\pi}{3}i} + 1\right)^3 = -3\sqrt{3}i.$$

□

3.7. Exercise 1.C.7: A Quasi-subspace Unclosed under Scalar Multiplication

Give an example of a nonempty subset U of $\mathbb{R}^2$ such that U is closed under addition and under taking additive inverses (meaning $-u \in U$ whenever $u \in U$), but U is not a subspace of $\mathbb{R}^2$.

Answer.

$$U = \{(0,0), (1,1), (-1,-1)\}.$$

□

3.8. Exercise 1.C.8: A Quasi-subspace Unclosed under Vector Addition

Give an example of a nonempty subset U of $\mathbb{R}^2$ such that U is closed under scalar multiplication, but U is not a subspace of $\mathbb{R}^2$.

Answer.

$$U = \{(0,y) : y \in \mathbb{R}\} \cup \{(x,0) : x \in \mathbb{R}\}.$$

□

3.9. Exercise 1.C.9: Periodic Functions is NOT a Subspace of $\mathbb{R}^{\mathbb{R}}$

A function $f : \mathbb{R} \to \mathbb{R}$ is called periodic if there exists a positive number p such that $f(x) = f(x+p)$ for all $x \in \mathbb{R}$. Is the set of periodic functions from $\mathbb{R}$ to $\mathbb{R}$ a subspace of $\mathbb{R}^{\mathbb{R}}$? Explain.

Lemma 3.1. *$f(x) = \cos(\pi x) + \cos(x)$ is not periodic.*

Proof. Suppose that $f(x) = \cos(\pi x) + \cos(x)$ is periodic, then there exists $\theta \in \mathbb{R}$ such that $\theta > 0$ and $f(x+\theta) = f(x)$ for all $x \in \mathbb{R}$. Pick $x = 0$, we have

$$f(\theta + 0) = \cos(\pi\theta) + \cos(\theta) = f(0) = \cos(0) + \cos(0) = 2. \tag{3.1}$$

Note that $-1 \leqslant \cos(\pi\theta) \leqslant 1$ and $-1 \leqslant \cos(\theta) \leqslant 1$. However, if $\cos(\pi\theta) < 1$ and $\cos(\theta) < 1$, then $\cos(\pi\theta) + \cos(\theta) < 1$. Thus (3.1) implies that

$$\begin{cases} \cos(\pi\theta) = 1 \\ \cos(\theta) = 1 \end{cases}.$$

This further implies that there exists some $m, n \in \mathbb{Z}$ such that,

$$\begin{cases} \pi\theta = 2m\pi \\ \theta = 2n\pi \end{cases}.$$

Since $\theta \neq 0$, dividing the above two equations yields

$$\pi = \frac{m}{n}.$$

This contradicts with the fact that π is irrational. □

Main Answer. No, because Lemma 3.1 gives an example of two periodic functions where their addition is not periodic. □

3.10. Exercise 1.C.10: The Intersection of 2 Subspaces is a Subspace

Suppose U_1 and U_2 are subspaces of V. Prove that the intersection $U_1 \cap U_2$ is a subspace of V.

Proof. We prove by verifying the three conditions for $U_1 \cap U_2$ being a subspace.

(1) The premise that U_1 is a subspace implies $0 \in U_1$. The premise that U_2 is a subspace implies $0 \in U_2$. Therefore $0 \in U_1 \cap U_2$.
(2) Pick $u, w \in U_1 \cap U_2$. The premise that U_1 is a subspace implies $u + w \in U_1$. The premise that U_2 is a subspace implies $u + w \in U_2$. Therefore $u + w \in U_1 \cap U_2$.
(3) Pick $a \in \mathbb{F}$ and $u \in U_1 \cap U_2$. The premise that U_1 is a subspace implies that $au \in U_1$. The premise that U_2 is a subspace implies $au \in U_2$. Therefore $au \in U_1 \cap U_2$.

□

3.11. Exercise 1.C.11: The Intersection of Subspaces is a Subspace

Prove that the intersection of every collection of subspaces of V is a subspace of V.

Proof. Pick set I and $(S_i)_{i \in I}$, a family of sets indexed over set I. Suppose that S_i is a subspace of V for all $i \in I$. We prove by verifying the three conditions for $\cap_{i \in I} S_i$ being a subspace.

(1) For all $i \in I$, the premise that S_i is a subspace implies that $0 \in S_i$. Therefore $0 \in \cap_{i \in I} S_i$.
(2) Pick $u, w \in \cap_{i \in I} S_i$. For all $i \in I$, the premise that S_i is a subspace implies $u + w \in S_i$. Therefore $u + w \in \cap_{i \in I} S_i$.
(3) Pick $a \in \mathbb{F}$ and $u \in \cap_{i \in I} S_i$. For all $i \in I$, the premise that U_i is a subspace implies that $au \in U_i$. Therefore $au \in \cap_{i \in I} S_i$.

□

3.12. Exercise 1.C.12: When the Union of 2 Subspaces is a Subspace

Prove that the union of two subspaces of V is a subspace of V if and only if one of the subspaces is contained in the other.

Proof. Pick two subspaces V_1, V_2 of V.

(1) If $V_1 \subset V_2$, then $V_1 \cup V_2 = V_2$. Since V_2 is a subspace, it follows that $V_1 \cup V_2$ is a subspace.
(2) Suppose that $V_1 \cup V_2$ is a subspace. To prove by contradiction, suppose that $V_1 \not\subset V_2$ and $V_2 \not\subset V_1$. Thus there exists $v_1 \in V_1$ such that $v_1 \notin V_2$, and there exists $v_2 \in V_2$ such that $v_2 \notin V_1$.

    We claim that $v_1 + v_2 \notin V_1$. Because if $v_1 + v_2$ was in V_1, the premise that V_1 is a subspace would imply that $v_2 = (v_1 + v_2) - v_1$ is also in V_1.

    We claim that $v_1 + v_2 \notin V_2$. Because if $v_1 + v_2$ was in V_2, the premise that V_2 is a subspace would imply that $v_1 = (v_1 + v_2) - v_2$ is also in V_2.

    Thus $v_1 + v_2 \notin V_1 \cup V_2$, contradicting with the premise that $V_1 \cup V_2$ is a subspace.

□

3.13. Exercise 1.C.13: When the Union of 3 Subspaces is a Subspace

Prove that the union of three subspaces of V is a subspace of V if and only if one of the subspaces contains the other two.

PROOF. Pick three subspaces V_1, V_2, V_3 of V.

(1) If $V_1 \subset V_3$ and $V_2 \subset V_3$, then $V_1 \cup V_2 \cup V_3 = V_3$. Since V_3 is a subspace, it follows that $V_1 \cup V_2 \cup V_3$ is a subspace.

(2) Suppose that $V_1 \cup V_2 \cup V_3$ is a subspace of V.

If $V_1 \subset V_2$, then $V_1 \cup V_2 \cup V_3$ becomes $V_2 \cup V_3$. By Exercise 1.C.12, it follows that either $V_2 \subset V_3$ or $V_3 \subset V_2$. Thus either V_3 or V_2 is the subspace that contains the other two.

If $V_2 \subset V_1$, then $V_1 \cup V_2 \cup V_3$ becomes $V_1 \cup V_3$. By Exercise 1.C.12, it follows that either $V_1 \subset V_3$ or $V_3 \subset V_1$. Thus either V_3 or V_1 is the subspace that contains the other two.

If $V_1 \not\subset V_2$ and $V_2 \not\subset V_1$, then there exists $u \in V_1$ and $w \in V_2$ such that $u \notin V_2$ and $w \notin V_1$.

(i) We claim that $V_1 \setminus (V_1 \cap V_2) \subset V_3$. Pick $x \in V_1 \setminus (V_1 \cap V_2) \subset V_3$. The premise that $V_1 \cup V_2 \cup V_3$ is a subspace implies that $x + w \in V_1 \cup V_2 \cup V_3$ and $2x + w \in V_1 \cup V_2 \cup V_3$. If $x + w$ was in V_1, then $w = (x + w) - x$ would be in V_1 which is a contradiction. If $2x + w$ was in V_1, then $w = (2x + w) - 2x$ would be in V_1 which is a contradiction. If $x + w$ was in V_2, then $x = (x + w) - w$ would be in V_2 which is a contradiction. If $2x + w$ was in V_2, then $2x = (2x + w) - w$ would be in V_2 which is a contradiction. Therefore $x + w \in V_3$ and $2x + w \in V_3$. And it follows that $x = (2x + w) - (x + w)$ is in V_3.

(ii) We claim that $V_2 \setminus (V_1 \cap V_2) \subset V_3$. Pick $y \in V_2 \setminus (V_1 \cap V_2) \subset V_3$. The premise that $V_1 \cup V_2 \cup V_3$ is a subspace implies that $u + y \in V_1 \cup V_2 \cup V_3$ and $u + 2y \in V_1 \cup V_2 \cup V_3$. If $u + y$ was in V_1, then $y = (u + y) - u$ would be in V_1 which is a contradiction. If $u + 2y$ was in V_1, then $2y = (u + 2y) - u$ would be in V_1 which is a contradiction. If $u + y$ was in V_2, then $u = (u + y) - y$ would be in V_2 which is a contradiction. If $u + 2y$ was in V_2, then $u = (u + 2y) - 2y$ would be in V_2 which is a contradiction. Therefore $u + y \in V_3$ and $u + 2y \in V_3$. And it follows that $y = (u + 2y) - (u + y)$ is in V_3.

(iii) We claim that $V_1 \cap V_2 \subset V_3$. Pick $z \in V_1 \cap V_2$. The premise that V_1 is a subspace implies that $u + z \in V_1$. If $u + z \in V_1 \cap V_2$, then $u = (u + z) - z \in V_2$ which is a contradiction. Thus $u + z \in V_1 \setminus (V_1 \cap V_2)$. By (i) we have $u + z \in V_3$. Also by (i) we have $u \in V_3$. The premise that V_3 is a subspace implies that $z = (u + z) - u$ is in V_3.

The above (i) and (iii) implies that $V_1 \subset V_3$. The above (ii) and (iii) implies that $V_2 \subset V_3$.

□

3.14. Exercise 1.C.14: An Example of Sums of Subspaces

Verify the assertion in Example 1.38: Suppose that $U = \{(x, x, y, y) \in \mathbb{F}^4 : x, y \in \mathbb{F}\}$ and W=$\{(x, x, x, y) \in \mathbb{F}^4 : x, y \in \mathbb{F}\}$. Then

$$U + W = \{(x, x, y, z) \in \mathbb{F}^4 : x, y, z \in \mathbb{F}\}.$$

PROOF. (1) Pick $x \in U + W$. Then there exists $u = (x_u, x_u, y_u, y_u) \in U$ and $w = (x_w, x_w, x_w, y_w) \in W$ such that $x_u, y_u, x_w, y_w \in \mathbb{F}$ and $x = u + w$. Thus $x = (x_u +$

$x_w, x_u + x_w, y_u + x_w, y_u + y_w)$, which shows that $x \in \{(x, x, y, z) \in \mathbb{F}^4 : x, y, z \in \mathbb{F}\}$. Therefore $U + W \subset \{(x, x, y, z) \in \mathbb{F}^4 : x, y, z \in \mathbb{F}\}$.

(2) Pick $v \in \{(x, x, y, z) \in \mathbb{F}^4 : x, y, z \in \mathbb{F}\}$. Then there exists $x, y, z \in \mathbb{F}$ such that $v = (x, x, y, z)$. By choosing $u = (x, x, y, y) \in U$ and $w = (0, 0, 0, z - y) \in W$, we have $u + w = v$. Thus $v \in U + W$. Therefore $\{(x, x, y, z) \in \mathbb{F}^4 : x, y, z \in \mathbb{F}\} \subset U + W$.

The above (1) and (2) imply that $U + W = \{(x, x, y, z) \in \mathbb{F}^4 : x, y, z \in \mathbb{F}\}$. □

3.15. Exercise 1.C.15: $U + U = U$ for Any Subspace U

Suppose U is a subspace of V. What is $U + U$?

Proof. We claim that $U + U = U$.

(1) Pick $z \in U + U$, then there exists $x, y \in U$ such that $z = x + y$. Since U is closed under addition, we have $x + y \in U$, i.e. $z \in U$. Thus $U + U \subset U$.

(2) Pick $u \in U$, then $u = 1u = (\frac{1}{2} + \frac{1}{2})u = \frac{1}{2}u + \frac{1}{2}u$, i.e. $u \in U + U$. Thus $U \subset U + U$.

The above (1) and (2) imply that $U + U = U$. □

3.16. Exercise 1.C.16: Commutativity of the Sum of Subspaces

Is the operation of addition on the subspaces of V commutative? In other words, if U and W are subspaces of V, is $U + W = W + U$?

Proof. We claim that $U + W = W + U$.

(1) Pick $v \in U + W$, then there exists $u \in U$ and $w \in W$ such that $v = u + w$. Since the addition on V is commutative, $u + w = w + u$, and this shows that $v = w + u$ is in $W + U$. Thus $U + W \subset W + U$.

(2) Pick $v \in W + U$, then there exists $w \in W$ and $u \in U$ such that $v = w + u$. Since the addition on V is commutative, $w + u = u + w$, and this shows that $v = u + w$ is in $U + W$. Thus $W + U \subset U + W$.

The above (1) and (2) imply that $U + W = W + U$. □

3.17. Exercise 1.C.17: Associativity of the Sum of Subspaces

Is the operation of addition on the subspaces of V associative? In other words, if U_1, U_2, U_3 are subspaces of V, is

$$(U_1 + U_2) + U_3 = U_1 + (U_2 + U_3)?$$

Proof. We claim that $(U_1 + U_2) + U_3 = U_1 + (U_2 + U_3)$.

(1) Pick $v \in (U_1 + U_2) + U_3$, then there exists $u_1, u_2, u_3 \in U$ such that $v = (u_1 + u_2) + u_3$. Since the addition on V is associative, $(u_1 + u_2) + u_3 = u_1 + (u_2 + u_3)$, and this shows that $v = u_1 + (u_2 + u_3)$ is in $U_1 + (U_2 + U_3)$. Thus $(U_1 + U_2) + U_3 \subset U_1 + (U_2 + U_3)$.

(2) Pick $v \in U_1 + (U_2 + U_3)$, then there exists $u_1, u_2, u_3 \in U$ such that $v = u_1 + (u_2 + u_3)$. Since the addition on V is associative, $u_1 + (u_2 + u_3) = (u_1 + u_2) + u_3$, and this shows that $v = (u_1 + u_2) + u_3$ is in $(U_1 + U_2) + U_3$. Thus $U_1 + (U_2 + U_3) \subset (U_1 + U_2) + U_3$.

The above (1) and (2) imply that $(U_1 + U_2) + U_3 = U_1 + (U_2 + U_3)$. □

3.18. Exercise 1.C.18: The Identity of Sum of Subspaces

Does the operation of addition on the subspaces of V have an additive identity? Which subspaces have additive inverses?

PROOF. We claim that the subspace $\{0\}$ is the additive identity, i.e. $U + \{0\} = U$ for all subspaces U.

(1) Pick $v \in U + \{0\}$, then there exists $u \in U$ such that $v = u + 0$, and this shows that $v = u$ is in U. Thus $U + \{0\} \subset U$.
(2) Pick $u \in U$, then $u = u + 0$ shows that $u \in U + \{0\}$. Thus $U \subset U + \{0\}$.

The above (1) and (2) imply that $U + \{0\} = U$.

We claim that for all subspaces U and W, $U + W = \{0\}$ implies $U = W = \{0\}$, which means that the subspaces that have additive inverses are only $\{0\}$ itself.

Pick subspaces U and W. Assume $U + W = \{0\}$. 1.39 of [**Axl14**] shows that $U \subset U + W$ and $W \subset U + W$. Thus $U \subset \{0\}$ and $W \subset \{0\}$. Note that the only two subsets of $\{0\}$ are $\varnothing$ and $\{0\}$. Also, the premise that U and W are subspaces implies $0 \in U$ and $0 \in W$, thus $U = W = \{0\}$. □

3.19. Exercise 1.C.19: There is NO Cancellation Law for Sums of Subspaces

Prove or give a counterexample: if U_1, U_2, W are subspaces of V such that

$$U_1 + W = U_2 + W,$$

then $U_1 = U_2$.

COUNTEREXAMPLE. Set $V = \mathbb{R}^2, U_1 = \mathbb{R}^2, U_2 = \{(x, 0) : x \in \mathbb{R}\}, W = \{(0, x) : x \in \mathbb{R}\}$. Then U_1, U_2, W are subspaces of V, $U_1 + W = \mathbb{R}^2$, $U_2 + W = \mathbb{R}^2$ but $U_1 \neq U_2$. □

3.20. Exercise 1.C.20: A Direct Sum of 2 Subspaces that Equals to $\mathbb{F}^4$

Suppose

$$U = \{(x, x, y, y) \in \mathbb{F}^4 : x, y \in \mathbb{F}\}.$$

Find a subspace W of $\mathbb{F}^4$ such that $\mathbb{F}^4 = U \oplus W$.

PROOF. We claim that the following W is a subspace of $\mathbb{F}^4$ such that $\mathbb{F}^4 = U \oplus W$:

$$W = \{(0, x, 0, y) \in \mathbb{F}^4 : x, y \in \mathbb{F}\}.$$

We first verify that W is a subspace of $\mathbb{F}^4$ by 1.34 of [**Axl14**].

(1) We can show that $0 \in W$ by choosing $x = 0, y = 0$ in $(0, x, 0, y)$.
(2) Suppose that $u, w \in W$ where $u = (0, x_u, 0, y_u)$ and w=$(0, x_w, 0, y_w)$. Then $u + w = (0, x_u + x_w, 0, y_u + y_w)$ shows that $u + w$ is in W.
(3) Pick $a \in \mathbb{F}$ and $u \in W$ where $u = (0, x_u, 0, y_u)$. Then $au = (0, ax_u, 0, ay_u)$ shows that au is in W.

We then verify that $U + W$ is a direct sum by 1.45 of [**Axl14**].

Pick $v \in U \cap W$. Then there exists $x, y, s, t \in \mathbb{F}$ such that $v = (x, x, y, y) = (0, s, 0, t)$, which implies that $x = 0, s = 0, y = 0, t = 0$, i.e. $v = 0$. Therefore $U + W$ is a direct sum.

Finally, we verify that $\mathbb{F}^4 = U \oplus W$.

(1) 1.39 of [**Axl14**] implies that $U \oplus W$ is a subspace of $\mathbb{F}^4$. Therefore $U \oplus W \subset \mathbb{F}^4$.
(2) Pick $v \in \mathbb{F}^4$ where $v = (a, b, c, d)$ and $a, b, c, d \in \mathbb{F}$. Define

$$u = (a, a, c, c), w = (0, b - a, 0, d - c).$$

Then $u \in U, w \in W$ and $v = u + w$, which shows that $v \in U \oplus W$. Thus $\mathbb{F}^4 \subset U \oplus W$.

□

3.21. Exercise 1.C.21: A Direct Sum of 2 Subspaces that Equals to $\mathbb{F}^5$

Suppose

$$U = \{(x, y, x + y, x - y, 2x) \in \mathbb{F}^5 : x, y \in \mathbb{F}\}.$$

Find a subspace W of $\mathbb{F}^5$ such that $\mathbb{F} = U \oplus W$.

Proof. We claim that the following W is a subspace of $\mathbb{F}^5$ such that $\mathbb{F}^4 = U \oplus W$:

$$W = \{(0, 0, x, y, z) \in \mathbb{F}^5 : x, y, z \in \mathbb{F}\}.$$

We first verify that W is a subspace of $\mathbb{F}^5$ by 1.34 of [**Axl14**].

(1) We can show that $0 \in W$ by choosing $x = 0, y = 0, z = 0$ in $(0, 0, x, y, z)$.
(2) Suppose that $u, w \in W$ where

$$u = (0, 0, x_u, y_u, z_u), w = (0, 0, x_w, y_w, z_w).$$

Then $u + w = (0, 0, x_u + x_w, y_u + y_w, z_u + z_w)$ shows that $u + w$ is in W.
(3) Pick $a \in \mathbb{F}$ and $u \in W$ where $u = (0, 0, x_u, y_u, z_u)$. Then

$$au = (0, 0, ax_u, ay_u, az_u)$$

shows that au is in W.

We then verify that $U + W$ is a direct sum by 1.45 of [**Axl14**].

Pick $v \in U \cap W$. Then there exists $x, y, a, b, c \in \mathbb{F}$ such that $v = (x, y, x + y, x - y, 2x) = (0, 0, a, b, c)$, which implies that $x = 0, y = 0, a = 0, b = 0, c = 0$, i.e. $v = 0$. Therefore $U + W$ is a direct sum.

Finally, we verify that $\mathbb{F}^5 = U \oplus W$.

(1) 1.39 of [**Axl14**] implies that $U \oplus W$ is a subspace of $\mathbb{F}^5$. Therefore $U + W \subset \mathbb{F}^5$.
(2) Pick $v \in \mathbb{F}^5$ where $v = (a, b, c, d, e)$ and $a, b, c, d, e \in \mathbb{F}$. Define $u = (a, b, a + b, a - b, 2a), w = (0, 0, c - a - b, d - a + b, e - 2a)$. Then $u \in U, w \in W$ and $v = u + w$, which shows that $v \in U + W$. Thus $\mathbb{F}^5 \subset U \oplus W$.

□

3.22. Exercise 1.C.22: A Direct Sum of 4 Subspaces that Equals to $\mathbb{F}^5$

Suppose

$$U = \{(x, y, x + y, x - y, 2x) \in \mathbb{F}^5 : x, y \in \mathbb{F}\}.$$

Find three subspaces W_1, W_2, W_3 of $\mathbb{F}^5$, none of which equals $\{0\}$, such that $\mathbb{F}^5 = U \oplus W_1 \oplus W_2 \oplus W_3$.

Proof. We claim that the following W_1, W_2, W_3 are subspaces of $\mathbb{F}^5$ such that $\mathbb{F}^5 = U \oplus W_1 \oplus W_2 \oplus W_3$:

$$W_1 = \{(0, 0, x, 0, 0) \in \mathbb{F}^5 : x \in \mathbb{F}\},$$
$$W_2 = \{(0, 0, 0, x, 0) \in \mathbb{F}^5 : x \in \mathbb{F}\},$$
$$W_3 = \{(0, 0, 0, 0, x) \in \mathbb{F}^5 : x \in \mathbb{F}\}.$$

Suppose that

$$0 = u + w_1 + w_2 + w_3 \tag{3.2}$$

for some $u = (x, y, x+y, x-y, 2x), w_1 = (0,0,a,0,0), w_2 = (0,0,0,b,0), w_3 = (0,0,0,0,c)$ where $x, y, a, b, c \in \mathbb{F}$. Then (3.2) is equivalent to the following system of linear equations.

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 1 & 0 \\ 2 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

It is now clear that the coefficient matrix is nonsingular, thus $x = y = a = b = c = 0$ is the only solution. By 1.44 of [**Axl14**], we conclude that $\mathbb{F}^5 = U \oplus W_1 \oplus W_2 \oplus W_3$. □

3.23. Exercise 1.C.23: There is NO Cancellation Law for Direct Sums

Prove or give a counterexample: if U_1, U_2, W are subspaces of V such that

$$V = U_1 \oplus W \text{ and } V = U_2 \oplus W,$$

then $U_1 = U_2$.

Counterexample. Set $V = \mathbb{R}^2, U_1 = \{(x,x) : x \in \mathbb{R}\}, U_2 = \{(-x,x) : x \in \mathbb{R}\}, W = \{(0,x) : x \in \mathbb{R}\}$. Then U_1, U_2, W are subspaces of V, $U_1 \oplus W = \mathbb{R}^2$, $U_2 \oplus W = \mathbb{R}^2$ but $U_1 \neq U_2$. □

3.24. Exercise 1.C.24: A Direct Sum Decomposition of $\mathbb{R}^{\mathbb{R}}$

A function $f : \mathbb{R} \to \mathbb{R}$ is called even if

$$f(-x) = f(x)$$

for all $x \in \mathbb{R}$. A function $f : \mathbb{R} \to \mathbb{R}$ is called odd if

$$f(-x) = -f(x)$$

for all $x \in \mathbb{R}$. Let U_e denote the set of real-valued even functions on $\mathbb{R}$ and let U_o denote the set of real-valued odd functions on $\mathbb{R}$. Show that $\mathbb{R}^{\mathbb{R}} = U_e \oplus U_o$.

Proof. We first verify that U_e and U_o are subspaces of $\mathbb{R}^{\mathbb{R}}$ by 1.34 of [**Axl14**].

(1) We can show that $0 \in U_e$ and $0 \in U_o$ because the $0(-x) = 0(x) = 0$ and $0(-x) = -0(x) = 0$ for all $x \in \mathbb{R}$.
(2) Suppose that $u_e, w_e \in U_e, u_o, w_o \in U_o$, then

$$\begin{aligned} (u_e + w_e)(-x) &= u_e(-x) + w_e(-x) \\ &= u_e(x) + w_e(x) \\ &= (u_e + w_e)(x). \end{aligned}$$

Thus $u_e + w_e \in U_e$. Also,

$$\begin{aligned} (u_o + w_o)(-x) &= u_o(-x) + w_o(-x) \\ &= -u_o(x) - w_o(x) \\ &= -(u_o(x) + w_o(x)) \\ &= -(u_o + w_o)(x). \end{aligned}$$

Thus $u_o + w_o \in U_o$.

(3) Pick $a \in \mathbb{F}$ and $u_e \in U_e, u_o \in U_o$, then

$$\begin{aligned}(au_e)(-x) &= a \cdot u_e(-x) \\ &= a \cdot u_e(x) \\ &= (au_e)(x).\end{aligned}$$

Thus $au_e \in U_e$. Also,

$$\begin{aligned}(au_o)(-x) &= a \cdot u_o(-x) \\ &= -a \cdot u_o(x) \\ &= -(au_o)(x).\end{aligned}$$

Thus $au_o \in U_o$.

We then verify that $U_e + U_o$ is a direct sum by 1.45 of [**Axl14**].

Pick $u \in U_e \cap U_o$, then $u(-x) = u(x) = -u(x)$ for all $x \in \mathbb{R}$. Thus $u(x) = 0$ for all $x \in \mathbb{R}$, which means that u is the 0 function.

Finally, we prove that $\mathbb{R}^{\mathbb{R}} = U_e \oplus U_o$.

(1) 1.39 of [**Axl14**] implies that $U_e \oplus U_o$ is a subspace of $\mathbb{R}^{\mathbb{R}}$. Therefore $U_e \oplus U_o \subset \mathbb{R}^{\mathbb{R}}$.

(2) Pick $f(x) \in \mathbb{R}^{\mathbb{R}}$. Define

$$f_e(x) = \frac{f(x) + f(-x)}{2}, f_o(x) = \frac{f(x) - f(-x)}{2}.$$

Then

$$\begin{aligned}f_e(-x) &= \frac{f(-x) + f(x)}{2} \\ &= \frac{f(x) + f(-x)}{2} \\ &= f_e(x),\end{aligned}$$

which implies that $f_e(x) \in U_e$. Also,

$$\begin{aligned}f_o(-x) &= \frac{f(-x) - f(x)}{2} \\ &= -\frac{f(x) - f(-x)}{2} \\ &= -f_o(x),\end{aligned}$$

which implies that $f_o(x) \in U_o$. And,

$$\begin{aligned}f_e(x) + f_o(x) &= \frac{f(x) + f(-x)}{2} + \frac{f(x) - f(-x)}{2} \\ &= -\frac{f(x) + f(-x) + f(x) - f(-x)}{2} \\ &= f(x),\end{aligned}$$

which shows that $f(x) \in U_e \oplus U_o$. Therefore $\mathbb{R}^{\mathbb{R}} \subset U_e \oplus U_o$.

□

CHAPTER 4

Span and Linear Independence

4.1. Exercise 2.A.1: Alternative Span from an Existing Span

Suppose v_1, v_2, v_3, v_4 spans V. Prove that the list

$$v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$$

also spans V.

Proof. Denote $w_1 = v_1 - v_2, w_2 = v_2 - v_3, w_3 = v_3 - v_4, w_4 = v_4$. Let

$$A = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

then A is nonsingular and

$$A^{-1} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Moreover,

$$A \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{bmatrix}, \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = A^{-1} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{bmatrix}.$$

(1) Pick $x \in \operatorname{span}(v_1, v_2, v_3, v_4)$, then there exists $x_1, x_2, x_3, x_4 \in \mathbb{F}$ such that

$$\begin{aligned} v &= \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} \\ &= \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{bmatrix} \\ &= \begin{bmatrix} x_1 & x_1 + x_2 & x_1 + x_2 + x_3 & x_1 + x_2 + x_3 + x_4 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{bmatrix}. \end{aligned}$$

Therefore $v \in \operatorname{span}(w_1, w_2, w_3, w_4)$.

(2) Pick $w \in \operatorname{span}(w_1, w_2, w_3, w_4)$, then there exists $y_1, y_2, y_3, y_4 \in \mathbb{F}$ such that

$$\begin{aligned} w &= \begin{bmatrix} y_1 & y_2 & y_3 & y_4 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{bmatrix} \\ &= \begin{bmatrix} y_1 & y_2 & y_3 & y_4 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} \\ &= \begin{bmatrix} y_1 & y_2 - y_1 & y_3 - y_2 & y_4 - y_3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}. \end{aligned}$$

Therefore $w \in \operatorname{span}(v_1, v_2, v_3, v_4)$.

The above (1) and (2) imply that $\operatorname{span}(w_1, w_2, w_3, w_4) = \operatorname{span}(v_1, v_2, v_3, v_4)$. □

4.2. Exercise 2.A.2: Examples of Linearly Independent Lists

Verify the assertions in Example 2.18.

(a) A list v of one vector $v \in V$ is linearly independent if and only if $v \neq 0$.
(b) A list of two vectors in V is linearly independent if and only if neither vector is a scalar multiple of the other.
(c) $(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0)$ is linearly independent in $\mathbb{F}^4$.
(d) The list $1, z, \ldots, z^m$ is linearly independent in $\mathcal{P}(\mathbb{F})$ for each nonnegative integer m.

Proof. (a) If the list v is linearly independent but $v = 0$, then $1v = 0$ is a nontrivial representation of 0 using v which is a contradiction. If $v \neq 0$ but the list v is not linearly independent, then there exists $a \in \mathbb{F}$ such that $av = 0$ and $a \neq 0$ which contradicts 1.30 of [**Axl14**].

(b) If a list of two vectors v_1, v_2 in V is linearly independent and assume that, without loss of generality, $v_1 = av_2$ for some $a \in \mathbb{F}$, then $1v_1 - av_2 = 0$ is a nontrivial representation of 0 using v_1, v_2 which is a contradiction. If a list of two vectors v_1, v_2 in V is not linearly independent, then there exist some $a_1, a_2 \in \mathbb{F}$ such that $a_1v_1 + a_2v_2 = 0$ and not both of a_1 and a_2 are zero. If $a_1 \neq 0$, then $v_1 = -\frac{a_2}{a_1}v_2$; if $a_1 = 0$, then $v_2 = -\frac{a_1}{a_2}v_1$.

(c) Consider the equation $x(1, 0, 0, 0) + y(0, 1, 0, 0) + z(0, 0, 1, 0) = 0$ where $x, y, z \in \mathbb{F}$. In matrix form,

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Since the coefficient matrix is nonsingular, the only solution is $x = y = z = 0$.

(d) Consider the function $f(z) = x_0 + x_1z + \cdots + x_{N-1}z^{N-1}$ where $N - 1 \geq 0$ and $x_0, x_1, \ldots, x_{N-1} \in \mathbb{F}$. Suppose $f(z) = 0$ for all $z \in \mathbb{F}$. Then, in particular, $f(1) =$

$0, f(2) = 0, \ldots, f(N) = 0$. In matrix form,

$$\begin{bmatrix} 1 & 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 2^2 & 2^3 & \cdots & 2^{N-1} \\ 1 & 3 & 3^2 & 3^3 & \cdots & 3^{N-1} \\ 1 & 4 & 4^2 & 4^3 & \cdots & 4^{N-1} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & N & N^2 & N^3 & \cdots & N^{N-1} \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{N-1} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Note that the above coefficient matrix is a Vandermonde matrix. Since all the numbers $1, 2, \ldots, N$ are distinct, its determinant is non-zero. Therefore the only solution is $x_0 = x_1 = \cdots = x_{N-1} = 0$.

□

4.3. Exercise 2.A.3: a Linearly Dependent List

Find a number t such that

$$(3, 1, 4), (2, -3, 5), (5, 9, t)$$

is not linearly independent in $\mathbb{R}^3$.

Proof. We claim that $t = 2$ makes the list not linearly independent in $\mathbb{R}^3$. This can be shown by

$$\begin{aligned} &-3 \cdot (3, 1, 4) + 2 \cdot (2, -3, 5) + 1 \cdot (5, 9, 2) \\ &= ((-9) + 4 + 5, (-3) + (-6) + 9, (-12) + 10 + 2) \\ &= 0. \end{aligned}$$

□

4.4. Exercise 2.A.4: Conditions to Linear Dependence

Verify the assertion in the second bullet point in Example 2.20: the list

$$(2, 3, 1), (1, -1, 2), (7, 3, c)$$

is linearly dependent in $\mathbb{F}^3$ if and only if $c = 8$.

Proof. Let

$$A = \begin{bmatrix} 2 & 1 & 7 \\ 3 & -1 & 3 \\ 1 & 2 & c \end{bmatrix}.$$

Then the list $(2, 3, 1), (1, -1, 2), (7, 3, c)$ is linearly dependent if and only if $\det(A) = 0$. Since $\det(A) = 40 - 5c$, it follows that $\det(A) = 0$ if and only if $c = 8$. □

4.5. Exercise 2.A.5: Difference between $\mathbb{C}$ and $\mathbb{R}$ as Scalar Fields

(a) Show that if we think of $\mathbb{C}$ as a vector space over $\mathbb{R}$, then the list $(1+i, 1-i)$ is linearly independent.
(b) Show that if we think of $\mathbb{C}$ as a vector space over $\mathbb{C}$, then the list $(1+i, 1-i)$ is linearly dependent.

PROOF. Suppose that there exists $a+bi, c+di \in \mathbb{C}$ such that $a,b,c,d \in \mathbb{R}$ and $(a+bi)(1+i)+(c+di)(1-i)=0$, then

$$(a-b+c+d)+(a+b-c+d)i=0,$$

Rewrite the above linear system in the matrix form,

$$\begin{bmatrix} 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

then its solutions are

$$\left\{ \begin{bmatrix} -y \\ x \\ x \\ y \end{bmatrix} : x, y \in \mathbb{R} \right\}.$$

(a) If we think of $\mathbb{C}$ as a vector space over $\mathbb{R}$, then the list is linearly independent because

$$\left\{ \begin{bmatrix} -y \\ x \\ x \\ y \end{bmatrix} : x, y \in \mathbb{R} \right\} \cap \left\{ \begin{bmatrix} x \\ 0 \\ y \\ 0 \end{bmatrix} : x, y \in \mathbb{R} \right\} = \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \right\}.$$

(b) If we think of $\mathbb{C}$ as a vector space over $\mathbb{C}$, then the list is linearly dependent because

$$\begin{bmatrix} -1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \in \left\{ \begin{bmatrix} -y \\ x \\ x \\ y \end{bmatrix} : x, y \in \mathbb{R} \right\}$$

.

□

4.6. Exercise 2.A.6: Constructing Another Independent List from a List

Suppose v_1, v_2, v_3, v_4 is linearly independent in V. Prove that the list

$$v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$$

is also linearly independent.

PROOF. Use the same notation as the proof of Exercise 2.A.1. If the list

$$v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$$

is linearly dependent, then there exists $a_1, a_2, a_3, a_4 \in \mathbb{F}$, such that

$$\begin{bmatrix} a_1 & a_2 & a_3 & a_4 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{bmatrix} = 0$$

and

$$\begin{bmatrix} a_1 & a_2 & a_3 & a_4 \end{bmatrix} \neq \begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}.$$

Therefore,

$$\begin{bmatrix} a_1 & a_2 & a_3 & a_4 \end{bmatrix} A \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = 0.$$

Since A is nonsingular,

$$\begin{bmatrix} a_1 & a_2 & a_3 & a_4 \end{bmatrix} A \neq \begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}.$$

Therefore it follows that the list v_1, v_2, v_3, v_4 is linearly dependent. □

4.7. Exercise 2.A.7: Constructing Another Independent List from a List

Prove or give a counterexample: If $v_1, v_2, \ldots, v_m$ is a linearly independent list of vectors in V, then

$$5v_1 - 4v_2, v_2, v_3, \ldots, v_m$$

is linearly independent.

Proof. We claim that the proposition is true. Define

$$A = \begin{bmatrix} 5 & -4 & 0 & \ldots & 0 \\ 0 & 1 & 0 & \ldots & 0 \\ 0 & 0 & 1 & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & 1 \end{bmatrix}.$$

Suppose that

$$5v_1 - 4v_2, v_2, v_3, \ldots, v_m$$

is linearly dependent, then there exists $a_1, a_2, a_3, \ldots, a_m \in \mathbb{F}$, such that

$$\begin{bmatrix} a_1 & a_2 & a_3 & \ldots & a_m \end{bmatrix} A \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \ldots \\ v_m \end{bmatrix} = 0$$

and

$$\begin{bmatrix} a_1 & a_2 & a_3 & \ldots & a_m \end{bmatrix} \neq \begin{bmatrix} 0 & 0 & 0 & \ldots & 0 \end{bmatrix}.$$

Since $\det(A) = 5$, the matrix A is nonsingular. Thus

$$\begin{bmatrix} a_1 & a_2 & a_3 & \ldots & a_m \end{bmatrix} A \neq \begin{bmatrix} 0 & 0 & 0 & \ldots & 0 \end{bmatrix}.$$

Therefore it follows that the list $v_1, v_2, \ldots, v_m$ is linearly dependent. □

4.8. Exercise 2.A.8: Linear Independent List Multiplied by Scalar

Prove or give a counterexample: If $v_1, v_2, \ldots, v_m$ is a linearly independent list of vectors in V and $\lambda \in \mathbb{F}$ with $\lambda \neq 0$, then $\lambda v_1, \lambda v_2, \ldots, \lambda v_m$ is linearly independent.

Proof. We claim that the proposition is true. Suppose that $\lambda v_1, \lambda v_2, \ldots, \lambda v_m$ is linearly dependent, then there exists $a_1, a_2, \ldots, a_m$, not all 0, such that

$$a_1 \lambda v_1 + a_2 \lambda v_2 + \cdots + a_m \lambda v_m = 0.$$

Since $\lambda \neq 0$, λ^{-1} exists in $\mathbb{F}$. Multiplying the above equation by λ^{-1} yields

$$a_1 v_1 + a_2 v_2 + \cdots + a_m v_m = 0.$$

Note that not all of $a_1, a_2, \ldots, a_m$ are zero. Therefore it follows that $v_1, v_2, \ldots, v_m$ is a linearly dependent list. □

4.9. Exercise 2.A.9: Addition of Two Linearly Independent Lists

Prove or give a counterexample: If $v_1, \ldots, v_m$ and $w_1, \ldots, w_m$ are linearly independent lists of vectors in V, then $v_1 + w_1, \ldots, v_m + w_m$ is linearly independent.

Counterexample. Set $V = \mathbb{R}^2, m = 2, v_1 = (0,1), v_2 = (1,0), w_1 = (0,-1), w_2 = (-1,0)$. Then v_1, v_2 is a linearly independent list, and w_1, w_2 is also a linearly independent list. But

$$v_1 + w_1 = (0,0)$$

and

$$v_2 + w_2 = (0,0)$$

is not a linearly independent list. □

4.10. Exercise 2.A.10: Adding a Vector to a Linearly Independent List

Suppose $v_1, \ldots, v_m$ is linearly independent in V and $w \in V$. Prove that if $v_1 + w, \ldots, v_m + w$ is linearly dependent, then $w \in \operatorname{span}(v_1, \ldots, v_m)$.

Proof. If $v_1 + w, \ldots, v_m + w$ is linearly dependent, then there exists $a_1, \ldots, a_m$, not all 0, such that

$$a_1(v_1 + w) + \cdots + a_m(v_m + w) = 0.$$

The above equation implies that

$$w = -\frac{a_1}{a_1 + \cdots + a_m} v_1 - \cdots - \frac{a_m}{a_1 + \cdots + a_m} v_m,$$

thus $w \in \operatorname{span}(v_1, \ldots, v_m)$. □

4.11. Exercise 2.A.11: Concatenating a Vector to an Independent List

Suppose $v_1, \ldots, v_m$ is linearly independent in V and $w \in V$. Show that $v_1, \ldots, v_m, w$ is linearly independent if and only if

$$w \notin \operatorname{span}(v_1, \ldots, v_m).$$

Proof.

(1) Suppose that $v_1, \ldots, v_m, w$ is linearly dependent, then there exists $a_1, \ldots, a_m, a_w \in \mathbb{F}$, not all 0, such that

$$a_1 v_1 + \cdots + a_m v_m + a_w w = 0.$$

If $a_w = 0$, then the above equation would imply that $v_1, \ldots, v_m$ is linearly dependent, which is a contradiction. Therefore $a_w \neq 0$ and the above equation implies that

$$w = -\frac{a_1}{a_w} v_1 - \cdots - \frac{a_m}{a_w} v_m.$$

Thus $w \in \operatorname{span}(v_1, \ldots, v_m)$.

(2) Suppose that $w \in \operatorname{span}(v_1, \ldots, v_m)$, then there exists $a_1, \ldots, a_m \in \mathbb{F}$ such that

$$w = a_1 v_1 + \cdots + a_m v_m.$$

The above equation implies that

$$a_1 v_1 + \cdots + a_m v_m - w = 0.$$

Thus $v_1, \ldots, v_m, w$ is linearly dependent.

□

4.12. Exercise 2.A.12: Possible Linearly Independent Lists in $\mathcal{P}_4(\mathbb{F})$

Explain why there does not exist a list of six polynomials that is linearly independent in $\mathcal{P}_4(\mathbb{F})$.

PROOF. It is obvious that the list $1, z, z^2, z^3, z^4$ spans $\mathcal{P}_4(\mathbb{F})$, which has length 5. By 2.23 of [**Axl14**], the length of every linearly independent list of vectors in $\mathcal{P}_4(\mathbb{F})$ is therefore less than or equal to 5. Thus there does not exist a list of 6 polynomials that is linearly independent in $\mathcal{P}_4(\mathbb{F})$. □

4.13. Exercise 2.A.13: Possible Spans of $\mathcal{P}_4(\mathbb{F})$

Explain why no list of four polynomials spans $\mathcal{P}_4(\mathbb{F})$.

PROOF. Exercise 2.A.2 (d) has already proved that the list $1, z, z^2, z^3, z^4$ is linearly independent in $\mathcal{P}_4(\mathbb{F})$, which has length 5. By 2.23 of [**Axl14**], the length of every spanning list of $\mathcal{P}_4(\mathbb{F})$ is greater than or equal to 5. Thus no list of 4 polynomials spans $\mathcal{P}_4(\mathbb{F})$. □

4.14. Exercise 2.A.14: Another Definition of Infinite-Dimensional

Prove that V is infinite-dimensional if and only if there is a sequence $v_1, v_2, \ldots$ of vectors in V such that $v_1, \ldots, v_m$ is linearly independent for every positive integer m.

PROOF. (1) Suppose that V is finite-dimensional, then there exists a list $u_1, \ldots, u_k$ in V that spans V. By 2.23 of [**Axl14**], the length of every linearly independent list of vectors in V is less than or equal to k. Therefore, if we pick any vector sequence $v_1, v_2, \ldots$ in V, $v_1, \ldots, v_{k+1}$ must be linearly dependent.

(2) Suppose that V is infinite-dimensional. We can construct, by the following steps, a sequence $v_1, v_2, \ldots$ of vectors in V such that $v_1, \ldots, v_m$ is linearly independent for every positive integer m:

Step (1): Pick any $v \in V$ such that $v \neq 0$, and let $v_1 = v$. Note that the list v_1 is linearly independent.

Step (n): Suppose that Step $(n-1)$ has already created a vector list $v_1, \ldots, v_{n-1}$ such that $v_1, \ldots, v_m$ is linearly independent for every positive integer $m \in \{1, \ldots, n-1\}$. Since V is infinite-dimensional, there exists $w \in V$ such that $w \notin \operatorname{span}(v_1, \ldots, v_{n-1})$. Set $v_n = w$, and we claim that $v_1, \ldots, v_n$ is linearly independent. To prove this, suppose that there exists $a_1, \ldots, a_n \in \mathbb{F}$, not all zero, such that

$$a_1 v_1 + \cdots + a_n v_n = 0.$$

If $a_n = 0$, then $a_1, \ldots, a_{n-1}$ would be linearly dependent, which is a contradiction. If $a_n \neq 0$, then

$$v_n = -\frac{a_1}{a_n} v_1 - \cdots - \frac{a_{n-1}}{a_n} v_{n-1},$$

which implies that $v_n \in \operatorname{span}(v_1, \ldots, u_{n-1})$, which is also a contradiction.

□

4.15. Exercise 2.A.15: $\mathbb{F}^\infty$ is Infinite-Dimensional

Prove that $\mathbb{F}^\infty$ is infinite-dimensional.

PROOF. Suppose that $\mathbb{F}^\infty$ is finite-dimensional, then there exists a list $v_1, \dots, v_k \in \mathbb{F}^\infty$ such that $\mathbb{F}^\infty = \operatorname{span}(v_1, \dots, v_k)$. But we may construct the following independent list of vectors in $\mathbb{F}^\infty$ which has length $k+1$:

$$\begin{aligned}
v_1 &= (1, 0, 0, \dots, 0, 0, 0, 0, \dots) \\
v_2 &= (0, 1, 0, \dots, 0, 0, 0, 0, \dots) \\
v_3 &= (0, 0, 1, \dots, 0, 0, 0, 0, \dots) \\
&\vdots \\
v_{k-1} &= (0, 0, 0, \dots, 1, 0, 0, 0, \dots) \\
v_k &= (0, 0, 0, \dots, 0, 1, 0, 0, \dots) \\
v_{k+1} &= (0, 0, 0, \dots, 0, 0, 1, 0, \dots)
\end{aligned}$$

This contradicts with 2.23 of [**Axl14**]. □

4.16. Exercise 2.A.16: $\mathbb{R}^{[0,1]}$ is Infinite-Dimensional

Prove that the real vector space of all continuous real-valued functions on the interval $[0, 1]$ is infinite-dimensional.

PROOF. Suppose that $\mathbb{R}^{[0,1]}$ is finite-dimensional, then there exists a list

$$v_1, \dots, v_k \in \mathbb{R}^{[0,1]}$$

such that $\mathbb{R}^{[0,1]} = \operatorname{span}(v_1, \dots, v_k)$. But we may construct the following independent list of vectors in $\mathbb{R}^{[0,1]}$ which has length $k+1$:

$$\begin{aligned}
v_1(x) &= 1 \text{ if } x = \frac{1}{k+2}; \text{ otherwise } 0 \\
v_2(x) &= 1 \text{ if } x = \frac{2}{k+2}; \text{ otherwise } 0 \\
v_3(x) &= 1 \text{ if } x = \frac{3}{k+2}; \text{ otherwise } 0 \\
&\vdots \\
v_{k-1}(x) &= 1 \text{ if } x = \frac{k-1}{k+2}; \text{ otherwise } 0 \\
v_k(x) &= 1 \text{ if } x = \frac{k}{k+2}; \text{ otherwise } 0 \\
v_{k+1}(x) &= 1 \text{ if } x = \frac{k+1}{k+2}; \text{ otherwise } 0.
\end{aligned}$$

This contradicts with 2.23 of [**Axl14**]. □

4.17. Exercise 2.A.17: Polynomials that Vanish at the Same Point

Suppose $p_0, p_1, \dots, p_m$ are polynomials in $\mathcal{P}_m(\mathbb{F})$ such that $p_j(2) = 0$ for each j. Prove that $p_0, p_1, \dots, p_m$ is not linearly independent in $\mathcal{P}_m(\mathbb{F})$.

PROOF. Denote $p_j(z) = a_{j0} + a_{j1}z + a_{j2}z^2 + \cdots + a_{jm}z^m$ for each j and let

$$A = \begin{bmatrix} a_{00} & a_{01} & a_{02} & \dots & a_{0m} \\ a_{00} & a_{01} & a_{02} & \dots & a_{0m} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m0} & a_{m1} & a_{m2} & \dots & a_{mm} \end{bmatrix}.$$

That $p_j(2) = 0$ for each j implies

$$A \begin{bmatrix} 2^0 \\ 2^1 \\ 2^2 \\ \vdots \\ 2^m \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Thus A is a singular matrix, and so is A^T.

Suppose there exists $s_0, s_1, \dots, s_m \in \mathbb{F}$ such that $s_0p_0 + s_1p_1 + \cdots + s_mp_m = 0$, then

$$\begin{bmatrix} a_{00} & a_{10} & \dots & a_{m0} \\ a_{01} & a_{11} & \dots & a_{m1} \\ \vdots & \vdots & \ddots & \vdots \\ a_{0m} & a_{1m} & \dots & a_{mm} \end{bmatrix} \begin{bmatrix} s_0 \\ s_1 \\ \vdots \\ s_m \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Note that the coefficient matrix of the above linear system is A^T, which we have shown is a singular matrix. So the linear system has a nonzero solution. We therefore conclude that the list $p_0, p_1, \dots, p_m$ is linearly dependent. $\square$

CHAPTER 5

Bases

5.1. Exercise 2.B.1: Vector Spaces that Have Exactly One Basis

Find all vector spaces that have exactly one basis.

Proof. We claim that no vector spaces over $\mathbb{F}$ (which is either $\mathbb{R}$ or $\mathbb{C}$) have exactly one basis.

Given a vector space V. Suppose that $v_1, \ldots, v_k$ is the only basis of V. Denote

$$v = \begin{bmatrix} v_1 \\ \vdots \\ v_k \end{bmatrix}.$$

Note that Av is also a basis of V for any nonsingular matrix A, then

$$Av = v, \text{ for all nonsingular } A.$$

In particular, take $A = 2I$. Then

$$(2I - I)v = 0,$$

which implies that $v = 0$. But the list 0 is not linearly dependent, contradicting with the presumption that $v_1, \ldots, v_k$ is a basis. □

5.2. Exercise 2.B.2: Examples and Counterexamples of Bases

Verify all the assertions in Example 2.28.

(a) The list $(1, 0, \ldots, 0), (0, 1, 0, \ldots, 0), \ldots, (0, \ldots, 0, 1)$ is a basis of $\mathbb{F}^n$, called the standard basis of $\mathbb{F}^n$.
(b) The list $(1, 2), (3, 5)$ is a basis of $\mathbb{F}^2$.
(c) The list $(1, 2, -4), (7, -5, 6)$ is linearly independent in $\mathbb{F}^3$ but is not a basis of $\mathbb{F}^3$ because it does not span $\mathbb{F}^3$.
(d) The list $(1, 2), (3, 5), (4, 13)$ spans $\mathbb{F}^2$ but is not a basis of $\mathbb{F}^2$ because it is not linearly independent.
(e) The list $(1, 1, 0), (0, 0, 1)$ is a basis of $\{(x, x, y) \in \mathbb{F}^3 : x, y \in \mathbb{F}\}$.
(f) The list $(1, -1, 0), (1, 0, -1)$ is a basis of

$$\{(x, y, z) \in \mathbb{F}^3 : x + y + z = 0\}.$$

(g) The list $1, z, \ldots, z^m$ is a basis of $\mathcal{P}_m(\mathbb{F})$.

Proof. (a) Pick $v = (v_1, v_2, \ldots, v_n) \in \mathbb{F}^n$ where $v_1, v_2, \ldots, v_n \in \mathbb{F}$. Then it is obvious that v can be written uniquely as

$$v_1(1, 0, \ldots, 0) + v_2(0, 1, 0, \ldots, 0) + \cdots + v_n(0, \ldots, 0, 1).$$

(b) Pick $v = (v_1, v_2) \in \mathbb{F}^2$. Let $v = x(1,2) + y(3,5)$ where $x, y \in \mathbb{F}$, which corresponds to the following linear system

$$\begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}.$$

Since the coefficient of the above linear system is nonsingular, there is exactly one solution $\begin{bmatrix} x & y \end{bmatrix}^T$ for every $\begin{bmatrix} v_1 & v_2 \end{bmatrix}^T$. Therefore $(1,2), (3,5)$ is a basis of $\mathbb{F}^2$.

(c) As shown in (1), there exists a linearly independent list of length 3 in $\mathbb{F}^3$, thus no lists of length 2 can span $\mathbb{F}^3$ by 2.23 of [**Axl14**].

(d) As shown in (1), there exists a list of length 2 that spans $\mathbb{F}^2$, thus no lists of length 3 can be linearly independent in $\mathbb{F}^2$ by 2.23 of [**Axl14**].

(e) Pick $v = (x, x, y) \in \mathbb{F}^3$ where $x, y \in \mathbb{F}$. Then it is obvious that v can be written uniquely as

$$x(1,1,0) + y(0,0,1).$$

(f) Denote $V = \{(x, y, z) \in \mathbb{F}^3 : x + y + z = 0\}$. Pick $v = (a, b, c) \in V$ then $c = -a - b$. Let $v = x(1, -1, 0) + y(1, 0, -1)$ where $x, y \in \mathbb{F}$, which corresponds to the following linear system

$$\begin{bmatrix} 1 & 1 \\ -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a \\ b \\ -a-b \end{bmatrix}.$$

Note that the above linear system is an overdetermined system, so it has either no solutions or a unique solution. Moreover, it can be shown that

$$\begin{bmatrix} -b \\ a+b \end{bmatrix}$$

is one possible solution. Therefore every $v \in V$ can be written uniquely using $(1, -1, 0), (1, 0, -1)$.

(g) Pick $v(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_m z^m \in \mathcal{P}_m(\mathbb{F})$ where $a_0, a_1, a_2, \ldots, a_m \in \mathbb{F}$. Then 4.7 of [**Axl14**] implies that v can be written uniquely as

$$a_0 \cdot 1 + a_1 \cdot z + a_2 \cdot z^2 + \cdots + a_m \cdot z^m.$$

□

5.3. Exercise 2.B.3: Find and Extend a Basis for a Subspace of $\mathbb{R}^5$

(a) Let U be the subspace of $\mathbb{R}^5$ defined by

$$U = \{(x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5 : x_1 = 3x_2 \text{ and } x_3 = 7x_4\}.$$

Find a basis of U.

(b) Extend the basis in part (a) to a basis of $\mathbb{R}^5$.

(c) Find a subspace W of $\mathbb{R}^5$ such that $\mathbb{R}^5 = U \oplus W$.

Proof. (a) Let

$$\begin{aligned} u_1 &= (3, 1, 0, 0, 0) \\ u_2 &= (0, 0, 7, 1, 0) \\ u_3 &= (0, 0, 0, 0, 1). \end{aligned}$$

We claim that the list u_1, u_2, u_3 is a basis of U. To prove this, pick $u = (x_1, x_2, x_3, x_4, x_5) \in U$, then it is obvious that u can be uniquely represented as

$$x_2 u_1 + x_4 u_2 + x_5 u_3.$$

(b) Let

$$\begin{aligned} v_1 &= (1,0,0,0,0) \\ v_2 &= (0,1,0,0,0) \\ v_3 &= (0,0,1,0,0) \\ v_4 &= (0,0,0,1,0) \\ v_5 &= (0,0,0,0,1). \end{aligned}$$

Then v_1, v_2, v_3, v_4, v_5 is a basis of $\mathbb{R}^5$. Thus the list

$$u_1, u_2, u_3, v_1, v_2, v_3, v_4, v_5$$

spans $\mathbb{R}^5$. The procedure of the proof of 2.31 of [**Axl14**] deletes v_2, v_4, v_5 and leaves v_1, v_3 unchanged, resulting in the following basis of $\mathbb{R}^5$ extended from the basis in part (a):

$$u_1, u_2, u_3, v_1, v_3.$$

(c) Let $W = \operatorname{span}(v_1, v_3)$. We claim that $\mathbb{R}^5 = U \oplus W$. To prove this, by 1.45 of [**Axl14**], we only need to show that

$$\mathbb{R}^5 = U + W \text{ and } U \cap W = \{0\}.$$

To prove the 1st equation above, suppose $v \in \mathbb{R}^5$. Then because the list u_1, u_2, u_3, v_1, v_3 spans $\mathbb{R}^5$, there exists $a_1, a_2, a_3, b_1, b_3 \in \mathbb{F}$ such that

$$v = a_1u_1 + a_2u_2 + a_3u_3 + b_1v_1 + b_3v_3.$$

In other words, we have $v = u + w$, where $u = (a_1u_1 + a_2u_2 + a_3u_3) \in U$ and $w = (b_1v_1 + b_3v_3) \in W$. Thus $v \in U + W$, completing the proof that $\mathbb{R}^5 = U + W$.

To show that $U \cap W = \{0\}$, suppose $v \in U \cap W$. Then there exist scalars $a_1, a_2, a_3, b_1, b_3 \in \mathbb{F}$ such that

$$v = a_1u_1 + a_2u_2 + a_3u_3 = b_1v_1 + b_3v_3.$$

Thus

$$a_1u_1 + a_2u_2 + a_3u_3 - b_1v_1 - b_3v_3 = 0.$$

Because u_1, u_2, u_3, v_1, v_3 is linearly independent, this implies that $a_1 = a_2 = a_3 = b_1 = b_3 = 0$. Thus $v = 0$, completing the proof that $U \cap W = \{0\}$.

□

5.4. Exercise 2.B.4: Find and Extend a Basis for a Subspace of $\mathbb{C}^5$

(a) Let U be the subspace of $\mathbb{C}^5$ defined by

$$U = \{(z_1, z_2, z_3, z_4, z_5) \in \mathbb{C}^5 : 6z_1 = x_2 \text{ and } z_3 + 2z_4 + 3z_5 = 0\}.$$

Find a basis of U.

(b) Extend the basis in part (a) to a basis of $\mathbb{C}^5$.

(c) Find a subspace W of $\mathbb{C}^5$ such that $\mathbb{C}^5 = U \oplus W$.

Proof. (a) Let

$$\begin{aligned} u_1 &= (1,6,0,0,0) \\ u_2 &= (0,0,-2,1,0) \\ u_3 &= (0,0,-3,0,1). \end{aligned}$$

We claim that the list u_1, u_2, u_3 is a basis of U. To prove this, pick $u = (z_1, z_2, z_3, z_4, z_5) \in U$, then it is obvious that u can be uniquely represented as

$$x_1 u_1 + x_4 u_2 + x_5 u_3.$$

(b) Let

$$\begin{aligned} v_1 &= (1, 0, 0, 0, 0) \\ v_2 &= (0, 1, 0, 0, 0) \\ v_3 &= (0, 0, 1, 0, 0) \\ v_4 &= (0, 0, 0, 1, 0) \\ v_5 &= (0, 0, 0, 0, 1). \end{aligned}$$

Then v_1, v_2, v_3, v_4, v_5 is a basis of $\mathbb{C}^5$. Thus the list

$$u_1, u_2, u_3, v_1, v_2, v_3, v_4, v_5$$

spans $\mathbb{C}^5$. The procedure of the proof of 2.31 of [**Axl14**] deletes v_2, v_4, v_5 and leaves v_1, v_3 unchanged, resulting in the following basis of $\mathbb{C}^5$ extended from the basis in part (a):

$$u_1, u_2, u_3, v_1, v_3.$$

(c) Let $W = \operatorname{span}(v_1, v_3)$. We claim that $\mathbb{C}^5 = U \oplus W$. To prove this, by 1.45 of [**Axl14**], we only need to show that

$$\mathbb{C}^5 = U + W \text{ and } U \cap W = \{0\}.$$

To prove the 1st equation above, suppose $v \in \mathbb{C}^5$. Then because the list u_1, u_2, u_3, v_1, v_3 spans $\mathbb{C}^5$, there exists $a_1, a_2, a_3, b_1, b_3 \in \mathbb{F}$ such that

$$v = a_1 u_1 + a_2 u_2 + a_3 u_3 + b_1 v_1 + b_3 v_3.$$

In other words, we have $v = u + w$, where $u = (a_1 u_1 + a_2 u_2 + a_3 u_3) \in U$ and $w = (b_1 v_1 + b_3 v_3) \in W$. Thus $v \in U + W$, completing the proof that $\mathbb{C}^5 = U + W$.

To show that $U \cap W = \{0\}$, suppose $v \in U \cap W$. Then there exist scalars $a_1, a_2, a_3, b_1, b_3 \in \mathbb{F}$ such that

$$v = a_1 u_1 + a_2 u_2 + a_3 u_3 = b_1 v_1 + b_3 v_3.$$

Thus

$$a_1 u_1 + a_2 u_2 + a_3 u_3 - b_1 v_1 - b_3 v_3 = 0.$$

Because u_1, u_2, u_3, v_1, v_3 is linearly independent, this implies that $a_1 = a_2 = a_3 = b_1 = b_3 = 0$. Thus $v = 0$, completing the proof that $U \cap W = \{0\}$.

□

5.5. Exercise 2.B.5: A Basis of $\mathcal{P}_3(\mathbb{F})$ without Degree-2 Polynomials

Prove or disprove: there exists a basis p_0, p_1, p_2, p_3 of $\mathcal{P}_3(\mathbb{F})$ such that none of the polynomials p_0, p_1, p_2, p_3 has degree 2.

Proof. We claim that the following list is a possible choice of such p_0, p_1, p_2, p_3:

$$\begin{aligned} p_0(z) &= 1 + z^3 \\ p_1(z) &= z + z^3 \\ p_2(z) &= z^2 + z^3 \\ p_3(z) &= z^3. \end{aligned}$$

To prove this, note that p_0, p_1, p_2, p_3 each has degree 3. Thus it only suffices to show that p_0, p_1, p_2, p_3 is a basis of $\mathcal{P}_3(\mathbb{F})$.

Pick $p(z) \in \mathcal{P}_3(\mathbb{F})$. As shown in Exercise 2.B.2, $1, z, z^2, z^3$ is a basis $\mathcal{P}_3(\mathbb{F})$. Thus there exists a unique list of scalars $a_0, a_1, a_2, a_3 \in \mathbb{F}$ such that

$$p(z) = a_0 + a_1 z + a_2 z^2 + a_3 z^3.$$

Let us denote

$$A = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

then A is a nonsingular matrix and

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Using this notation, we have

$$\begin{bmatrix} p_0 \\ p_1 \\ p_2 \\ p_3 \end{bmatrix} = A \begin{bmatrix} 1 \\ z \\ z^2 \\ z^3 \end{bmatrix},$$

and

$$A^{-1} \begin{bmatrix} p_0 \\ p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} 1 \\ z \\ z^2 \\ z^3 \end{bmatrix}.$$

Therefore

$$\begin{aligned} p(z) &= \begin{bmatrix} a_0 & a_1 & a_2 & a_3 \end{bmatrix} \begin{bmatrix} 1 \\ z \\ z^2 \\ z^3 \end{bmatrix} \\ &= \left(\begin{bmatrix} a_0 & a_1 & a_2 & a_3 \end{bmatrix} A^{-1} \right) \begin{bmatrix} p_0 \\ p_1 \\ p_2 \\ p_3 \end{bmatrix}. \end{aligned}$$

The above equation implies that p_0, p_1, p_2, p_3 spans $\mathcal{P}_3(\mathbb{F})$.

To show that p_0, p_1, p_2, p_3 is linearly independent, suppose $b_0, b_1, b_2, b_3 \in \mathbb{F}$ are such that

$$0 = \begin{bmatrix} b_0 & b_1 & b_2 & b_3 \end{bmatrix} \begin{bmatrix} p_0 \\ p_1 \\ p_2 \\ p_3 \end{bmatrix}.$$

Then the above equation implies that

$$0 = \left(\begin{bmatrix} b_0 & b_1 & b_2 & b_3 \end{bmatrix} A \right) \begin{bmatrix} 1 \\ z \\ z^2 \\ z^3 \end{bmatrix}.$$

Since $1, z, z^2, z^3$ is linearly independent, this implies that

$$0 = \begin{bmatrix} b_0 & b_1 & b_2 & b_3 \end{bmatrix} A.$$

Since A is nonsingular, this further implies that

$$0 = \begin{bmatrix} b_0 & b_1 & b_2 & b_3 \end{bmatrix}.$$

Thus p_0, p_1, p_2, p_3 is linearly independent and hence is a basis of $\mathcal{P}_3(\mathbb{F})$. □

5.6. Exercise 2.B.6: Constructing Another Basis from a Basis

Suppose v_1, v_2, v_3, v_4 is a basis of V. Prove that

$$v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$$

is also a basis of V.

Proof. Denote

$$\begin{aligned} u_1 &= v_1 + v_2 \\ u_2 &= v_2 + v_3 \\ u_3 &= v_3 + v_4 \\ u_4 &= v_4. \end{aligned}$$

Pick $v \in V$, then there exists a unique list of scalars $a_1, a_2, a_3, a_4 \in \mathbb{F}$ such that

$$v = a_1 v_1 + a_2 v_2 + a_3 v_3 + a_4 v_4.$$

Let us denote

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

then A is a nonsingular matrix and

$$A^{-1} = \begin{bmatrix} 1 & -1 & 1 & -1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Using this notation, we have

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = A \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix},$$

and

$$A^{-1} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}.$$

Therefore

$$\begin{aligned} v &= \begin{bmatrix} a_0 & a_1 & a_2 & a_3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} \\ &= \left(\begin{bmatrix} a_0 & a_1 & a_2 & a_3 \end{bmatrix} A^{-1} \right) \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix}. \end{aligned}$$

The above equation implies that u_1, u_2, u_3, u_4 spans V.
To show that u_1, u_2, u_3, u_4 is linearly independent, suppose $b_1, b_2, b_3, b_4 \in \mathbb{F}$ are such that

$$0 = \begin{bmatrix} b_1 & b_2 & b_3 & b_4 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix}.$$

Then the above equation implies that

$$0 = \left(\begin{bmatrix} b_1 & b_2 & b_3 & b_4 \end{bmatrix} A \right) \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}.$$

Since v_1, v_2, v_3, v_4 is linearly independent, this implies that

$$0 = \begin{bmatrix} b_1 & b_2 & b_3 & b_4 \end{bmatrix} A.$$

Since A is nonsingular, this further implies that

$$0 = \begin{bmatrix} b_1 & b_2 & b_3 & b_4 \end{bmatrix}.$$

Thus u_1, u_2, u_3, u_4 is linearly independent and hence is a basis of V. □

5.7. Exercise 2.B.7: Constructing a Counterexample

Prove or give a counterexample: If v_1, v_2, v_3, v_4 is a basis of V and U is a subspace of V such that $v_1, v_2 \in U$ and $v_3 \notin U$ and $v_4 \notin U$, then v_1, v_2 is a basis of U.

Counterexample. Take $V = \mathbb{R}^4$ and define

$$\begin{aligned} v_1 &= (2, 0, 1, 0), \\ v_2 &= (0, 0, 0, 1), \\ v_3 &= (0, 3, \frac{1}{2}, 1), \\ v_4 &= (3, 0, 1, \frac{1}{2}). \end{aligned}$$

We claim that v_1, v_2, v_3, v_4 is a basis of V, which can be shown by the following determinant calculation:

$$\det \begin{pmatrix} 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 3 & \frac{1}{2} & 1 \\ 3 & 0 & 1 & \frac{1}{2} \end{pmatrix} = -3.$$

Define

$$U = \{(2x, y, x, z) \in \mathbb{R}^4 : x, y, z \in \mathbb{R}\}.$$

We claim that U is a subspace of V. To show this, first $0 = (2 \cdot 0, 0, 0, 0)$ is in U. Suppose $u = (2x, y, x, z) \in U$ and $w = (2a, b, a, c) \in U$, we can check that $u + w = (2x + 2a, y + b, x + a, z + c) = (2(x + a), y + b, x + a, z + c)$ is in U. Suppose $f \in \mathbb{F}$ and $u = (2x, y, x, z) \in U$, we can check that $fu = (2fx, y, fx, z) = (2(fx), y, fx, z)$ is also in U.

Further, it is clear that $v_1, v_2 \in U, v_3 \notin U, v_4 \notin U$.

Finally, since $(0, 1, 0, 0) \in U$ cannot be written as a linear combination of v_1 and v_2, we conclude that v_1, v_2 is not a basis of U. □

5.8. Exercise 2.B.8: Basis Obtained from a Direct Sum

Suppose U and W are subspaces of V such that $V = U \oplus W$. Suppose also that $u_1, \ldots, u_m$ is a basis of U and $w_1, \ldots, w_n$ is a basis of W. Prove that

$$u_1, \ldots, u_m, w_1, \ldots, w_n$$

is a basis of V.

Proof. (1) We first prove that $u_1, \ldots, u_m, w_1, \ldots, w_n$ spans V. Pick $v \in V$. By the definition of direct sum, v can be written in the only one way as a sum $u + w$, where $u \in U$ and $w \in W$. By the definition of basis, u can be written as $u = a_1u_1 + \cdots + a_mu_m$ and w can be written as $w = b_1w_1 + \cdots + b_nw_n$ where $a_1, \ldots, a_m, b_1, \ldots, b_n \in \mathbb{F}$. Taken together,

$$v = a_1u_1 + \cdots + a_mu_m + b_1w_1 + \cdots + b_nw_n.$$

(2) We then prove that $u_1, \ldots, u_m, w_1, \ldots, w_n$ is linearly independent. Suppose there exist $x_1, \ldots, x_m, y_1, \ldots, y_n \in \mathbb{F}$ such that

$$x_1u_1 + \cdots + x_mu_m + y_1w_1 + \cdots + y_nw_n = 0.$$

Then

$$x_1u_1 + \cdots + x_mu_m = -y_1w_1 - \cdots - y_nw_n.$$

Denote $p = x_1u_1 + \cdots + x_mu_m$ and $q = -y_1w_1 - \cdots - y_nw_n$, then $p \in U \cap W$ and $q \in U \cap W$. By 1.45 of [**Axl14**], $p = q = 0$.

Further, since $u_1, \ldots, u_m$ is linearly independent, $p = 0$ implies that all of $x_1, \ldots, x_m$ are 0. Similarly, since $w_1, \ldots, w_n$ is linearly independent, $q = 0$ implies that all of $y_1, \ldots, y_n$ are 0.

□

CHAPTER 6

Dimension

6.1. Exercise 2.C.1: A Space with the Same Dimension of its Subspace

Suppose V is finite-dimensional and U is a subspace of V such that $\dim U = \dim V$. Prove that $U = V$.

PROOF. Suppose $\dim U = \dim V = n$ and $u_1, \ldots, u_n$ is a basis of U. Since $u_1, \ldots, u_n$ is a linearly independent list in V with length $\dim V$, 2.39 of [**Axl14**] implies that $u_1, \ldots, u_n$ is a basis of V. Hence

$$U = \operatorname{span}(u_1, \ldots, u_n) = V.$$

□

6.2. Exercise 2.C.2: Subspaces of $\mathbb{R}^2$

Show that the subspaces of $\mathbb{R}^2$ are precisely $\{0\}$, $\mathbb{R}^2$, and all lines in $\mathbb{R}^2$ through the origin.

PROOF. Pick a subspace U of $\mathbb{R}^2$, then $\dim U \in \{0, 1, 2\}$ by 2.39 of [**Axl14**].

(1) If $\dim U = 0$, then $U = \{0\}$.
(2) If $\dim U = 1$, then there exists a vector $v = (x, y) \in \mathbb{R}^2$ such that
$$U = \operatorname{span}(v).$$
Hence, geometrically, U can be viewed as a line in $\mathbb{R}^2$ passing through points $(0, 0)$ and (x, y).
(3) If $\dim U = 2$, then Exercise 2.C.1 implies that $U = \mathbb{R}^2$.

□

6.3. Exercise 2.C.3: Subspaces of $\mathbb{R}^3$

Show that the subspaces of $\mathbb{R}^3$ are precisely $\{0\}$, $\mathbb{R}^3$, all lines in $\mathbb{R}^3$ through the origin, and all planes in $\mathbb{R}^3$ through the origin.

PROOF. Pick a subspace U of $\mathbb{R}^3$, then $\dim U \in \{0, 1, 2, 3\}$ by 2.39 of [**Axl14**].

(1) If $\dim U = 0$, then $U = \{0\}$.
(2) If $\dim U = 1$, then there exists a vector $v = (x, y, z) \in \mathbb{R}^3$ such that
$$U = \operatorname{span}(v).$$
Hence, geometrically, U can be viewed as a line in $\mathbb{R}^3$ passing through points $(0, 0, 0)$ and (x, y, z).
(3) If $\dim U = 2$, then there exists a vector $v = (v_1, v_2, v_3), w = (w_1, w_2, w_3)$ in $\mathbb{R}^3$ such that
$$U = \operatorname{span}(v, w).$$
Geometrically, since the list v, w is linearly independent, v and w are not on the same line in $\mathbb{R}^3$. Hence U is a plane in $\mathbb{R}^3$ that passes the following three points: $(v_1, v_2, v_3), (w_1, w_2, w_3)$ and $(0, 0, 0)$.

(4) If $\dim U = 3$, then Exercise 2.C.1 implies that $U = \mathbb{R}^3$.

□

6.4. Exercise 2.C.4: Find and Extend a Basis for a Subspace of $\mathcal{P}_4(\mathbb{F})$

(a) Let $U = \{p \in \mathcal{P}_4(\mathbb{F}) : p(6) = 0\}$. Find a basis of U.
(b) Extend the basis in part (a) to a basis of $\mathcal{P}_4(\mathbb{F})$.
(c) Find a subspace W of $\mathcal{P}_4(\mathbb{F})$ such that $\mathcal{P}_4(\mathbb{F}) = U \oplus W$.

Proof. (a) We first verify that U is a subspace of $\mathcal{P}_4$. First $0 \in U$ since $0(6) = 0$. If $p, q \in U$, then $p + q \in U$ because $(p+q)(6) = p(6) + q(6) = 0$. If $a \in \mathbb{F}$ and $p \in U$, then $ap \in U$ because $(ap)(6) = ap(6) = 0$.

We claim that the following list is a basis of U:

$$z - 6, z^2 - 6^2, z^3 - 6^3, z^4 - 6^4.$$

Clearly each of the above polynomials is in U. Suppose $a, b, c, d \in \mathbb{F}$ and

$$a(z-6) + b(z^2 - 6^2) + c(z^3 - 6^3) + d(z^4 - 6^4) = 0$$

for every $z \in \mathbb{F}$. Without explicitly expanding the left side of the equation above, we can see that the left side has a dz^4 term. Because the right side has no z^4 term, this implies that $d = 0$. Because $d = 0$, we see that the left side has a cz^3 term, which implies that $c = 0$. Similarly, $b = a = 0$.

Thus the equation above implies that $a = b = c = d = 0$. Hence the list is linearly independent in U. Thus $\dim U \geqslant 4$. Considering U as a subspace of $\mathcal{P}_4(\mathbb{F})$, which has dimension 5, we conclude that

$$4 \leqslant \dim U \leqslant 5.$$

But Exercise 2.A.17 has shown that any list of vectors in U that has length 5 is not linearly independent, which means that $\dim U \neq 5$. Thus $\dim U = 4$ and 2.39 of [**Axl14**] implies that the list is a basis of U.

(b) We claim that the following list is a basis of $\mathcal{P}_4(\mathbb{F})$:

$$1, z - 6, z^2 - 6^2, z^3 - 6^3, z^4 - 6^4.$$

Clearly each of the above polynomials is in $\mathcal{P}_4(\mathbb{F})$. Suppose

$$a_0, a_1, a_2, a_3, a_4 \in \mathbb{F}$$

and

$$a_0 + a_1(z-6) + a_2(z^2 - 6^2) + a_3(z^3 - 6^3) + a_4(z^4 - 6^4) = 0$$

for every $z \in \mathbb{F}$. Without explicitly expanding the left side of the equation above, we can see that the left side has a $a_4 z^4$ term. Because the right side has no z^4 term, this implies that $a_4 = 0$. Because $a_4 = 0$, we see that the left side has a $a_3 z^3$ term, which implies that $a_3 = 0$. Similarly, $a_2 = a_1 = 0$. Because $a_4 = a_3 = a_2 = a_1 = 0$, we can also conclude that $a_0 = 0$.

Hence the list is linearly independent in $\mathcal{P}_4(\mathbb{F})$. Since $\dim \mathcal{P}_4(\mathbb{F}) = 5$, 2.39 of [**Axl14**] implies that the list is a basis of $\mathcal{P}_4$.

(c) Define

$$W \equiv \{a_0 : a_0 \in \mathbb{F}\}$$

and we claim that $\mathcal{P}_4(\mathbb{F}) = U \oplus W$.

To prove this, by 1.45 of [**Axl14**], we only need to show that

$$\mathcal{P}_4(\mathbb{F}) = U + W \text{ and } U \cap W = \{0\}.$$

To prove the first equation above, suppose $v \in \mathcal{P}_4(\mathbb{F})$. Then, because the list $1, z-6, z^2-6^2, z^3-6^3, z^4-6^4$ spans $\mathcal{P}_4$, there exists $a_0, a_1, a_2, a_3, a_4 \in \mathbb{F}$ such that

$$v = \underbrace{a_0}_{w} + \underbrace{a_1(z-6) + a_2(z^2-6^2) + a_3(z^3-6^3) + a_4(z^4-6^4)}_{u}.$$

In other words, we have $v = u + w$, where $u \in U$ and $w \in W$ are defined as above. Thus $v \in U + W$, completing the proof that $\mathcal{P}_4(\mathbb{F}) = U + W$.

To show that $U \cap W = \{0\}$, suppose $v \in U \cap W$. Then there exist scalars $a_0, a_1, a_2, a_3, a_4 \in \mathbb{F}$ such that

$$v = a_0 = a_1(z-6) + a_2(z^2-6^2) + a_3(z^3-6^3) + a_4(z^4-6^4).$$

Thus

$$a_0 - a_1(z-6) - a_2(z^2-6^2) - a_3(z^3-6^3) - a_4(z^4-6^4) = 0.$$

Because because the list $1, z-6, z^2-6^2, z^3-6^3, z^4-6^4$ is linearly independent, this implies that $a_0 = a_1 = a_2 = a_3 = a_4 = 0$. Thus $v = 0$, completing the proof that $U \cap W = \{0\}$.

□

6.5. Exercise 2.C.9: $\dim \operatorname{span}(v_1 + w, \ldots, v_m + w) \geqslant m - 1$

Suppose $v_1, \ldots, v_m$ is linearly independent in V and $w \in V$. Prove that

$$\dim \operatorname{span}(v_1 + w, \ldots, v_m + w) \geqslant m - 1.$$

Proof. Define

$$\begin{aligned}
U &\equiv \operatorname{span}(v_1 + w, \ldots, v_m + w),\\
w_1 &\equiv (v_1 + w) - (v_2 + w) = v_1 - v_2,\\
w_2 &\equiv (v_2 + w) - (v_3 + w) = v_2 - v_3,\\
&\vdots\\
w_{m-1} &\equiv (v_{m-1} + w) - (v_m + w) = v_{m-1} - v_m,\\
W &\equiv \operatorname{span}(w_1, \ldots, w_{m-1}).
\end{aligned}$$

Then W is a subspace of U.
Suppose $a_1, \ldots, a_{m-1} \in \mathbb{F}$ and $a_1 w_1 + \cdots + a_{m-1} w_{m-1} = 0$, then

$$\begin{bmatrix} a_1 & a_2 & \ldots & a_{m-1} \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 & \ldots & 0 & 0 \\ 0 & 1 & -1 & \ldots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \ldots & 1 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_{m-1} \\ v_m \end{bmatrix} = 0.$$

Since $v_1, \ldots, v_m$ is linearly independent, the above equation implies

$$\begin{bmatrix} a_1 & a_2 & \ldots & a_{m-1} \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 & \ldots & 0 & 0 \\ 0 & 1 & -1 & \ldots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \ldots & 1 & -1 \end{bmatrix} = 0.$$

Transposing the above equation gives the following linear system:

$$\begin{bmatrix} 1 & 0 & \dots & 0 \\ -1 & 1 & \dots & 0 \\ 0 & -1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \\ 0 & 0 & \dots & -1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_{m-1} \end{bmatrix} = 0.$$

Denote $E_{i,j}$ as the elementary matrix produced by adding row i of the $m \times m$ identity matrix to row j, then left multiplication by $E_{m-1,m}E_{m-2,m-1} \dots E_{2,3}E_{1,2}$ gives

$$\begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \\ 0 & 0 & \dots & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_{m-1} \end{bmatrix} = 0.$$

Hence $a_1 = a_2 = \dots = a_{m-1} = 0$ and $w_1, \dots, w_{m-1}$ is linearly independent. It is clear that $w_1, \dots, w_{m-1}$ spans W. Hence $w_1, \dots, w_{m-1}$ is a basis of W. Therefore

$$\dim W = m - 1.$$

Finally, 2.38 of [**Axl14**] implies

$$\dim U \geqslant \dim W = m - 1.$$

□

CHAPTER 7

The Vector Space of Linear Maps

7.1. Exercise 3.A.1: Conditions for a Mapping to be Linear

Suppose $b, c \in \mathbb{R}$. Define $T : \mathbb{R}^3 \to \mathbb{R}^2$ by

$$T(x, y, z) = (2x - 4y + 3z + b, 6x + cxyz).$$

Show that T is linear if and only if $b = c = 0$.

PROOF. (1) Suppose $b = c = 0$, then

$$T(x, y, z) = (2x - 4y + 3z, 6x).$$

Pick $u = (u_1, u_2, u_3), v = (v_1, v_2, v_3)$, then

$$\begin{aligned} T(u + v) &= (2(u_1 + v_1) - 4(u_2 + v_2) + 3(u_3 + v_3), 6(u_1 + v_1)) \\ &= (2u_1 + 2v_1 - 4u_2 - 4v_2 + 3u_3 + 3v_3, 6u_1 + 6v_1) \end{aligned}$$

and

$$\begin{aligned} Tu + Tv &= (2u_1 - 4u_2 + 3u_3, 6u_1) + (2v_1 - 4v_2 + 3v_3, 6v_1) \\ &= (2u_1 - 4u_2 + 3u_3 + 2v_1 - 4v_2 + 3v_3, 6u_1 + 6v_1). \end{aligned}$$

Hence $T(u + v) = Tu + Tv$.

Pick $\lambda \in \mathbb{F}$, then

$$\begin{aligned} T(\lambda v) &= T(\lambda u_1, \lambda u_2, \lambda u_3) \\ &= (2\lambda u_1 - 4\lambda u_2 + 3\lambda u_3, 6\lambda u_1) \end{aligned}$$

and

$$\begin{aligned} \lambda T(v) &= \lambda(2u_1 - 4u_2 + 3u_3, 6u_1) \\ &= (2\lambda u_1 - 4\lambda u_2 + 3\lambda u_3, 6\lambda u_1). \end{aligned}$$

Hence $T(\lambda v) = \lambda T(v)$.

(2) By definition, if T is linear, then $T(u + v) = Tu + Tv$ for all $u, v \in \mathbb{R}^3$. In particular, take $u = (1, 1, 1), v = (1, 1, 1)$. Then

$$\begin{aligned} T(u + v) &= (2 \cdot 2 - 4 \cdot 2 + 3 \cdot 2 + b, 6 \cdot 2 + c \cdot 2 \cdot 2 \cdot 2) \\ &= (2 + b, 12 + 8c) \\ &= Tu + Tv \\ &= (2 - 4 + 3 + b + 2 - 4 + 3 + b, 6 + c + 6 + c) \\ &= (2 + 2b, 12 + 2c) \end{aligned}$$

Hence $b = 2b$ and $8c = 2c$, we thus conclude that $b = c = 0$.

□

7.2. Exercise 3.A.5: $\mathcal{L}(V, W)$ is a vector space

Prove the assertion in 3.7 of [**Axl14**]: $\mathcal{L}(V, W)$ is a vector space.

Proof. Define $\mathbf{0}$ as the linear map in $\mathcal{L}(V, W)$ such that

$$\mathbf{0}(v) = 0 \quad \forall v \in V.$$

Pick $R, S, T \in \mathcal{L}(V, W), v \in V$ and $a, b \in \mathbb{F}$.

(1) Commutativity: The commutativity of vector space W implies $Sv + Tv = Tv + Sv$. Hence $S + T = T + S$.
(2) Associativity: The associativity of vector space W implies

$$(Rv + Sv) + Tv = Rv + (Sv + Tv)$$

and

$$(ab)Sv = a(bSv).$$

Hence $(R + S) + T = R + (S + T)$ and $(ab)S = a(bS)$.
(3) Additive identity: It is clear that $T + \mathbf{0} = T$.
(4) Additive inverse: Define $-T$ as the linear map in $\mathcal{L}(V, W)$ such that

$$-T(v) = -(Tv) \quad \forall v \in V.$$

Then $T + (-T) = \mathbf{0}$.
(5) Multiplicative identity: The multiplicative identity of vector space W implies $(1T)(v) = 1(Tv) = Tv$. Hence $1T = T$.
(6) Distributive properties: The distributive properties of W implies

$$a(Sv + Tv) = aSv + aTv$$

and

$$(a + b)Sv = aSv + bSv.$$

Hence $a(S + T) = aS + aT$ and $(a + b)S = aS + bS$.

□

7.3. Exercise 3.A.6: Algebraic Properties of Products of Linear Maps

Prove the assertions in 3.9 of [**Axl14**]: The associativity, identity and distributive properties of Products of Linear Maps.

Proof. (1) Associativity: pick linear maps T_1, T_2 and T_3 such that T_3 maps into the domain of T_2, and T_2 maps into the domain of T_1. Then for any v in the domain of T_3,

$$((T_1T_2)T_3)(v) = (T_1T_2)(T_3(v)) = T_1(T_2(T_3(v))) = T_1(T_2T_3)(v).$$

Hence

$$(T_1T_2)T_3 = T_1(T_2T_3).$$

(2) Identity: Pick $T \in \mathcal{L}(V, W)$. Define I_V as the identity map on V and I_W as the identity map on W. Then for any $v \in V$,

$$(TI_V)(v) = T(I_V(v)) = T(v) = I_W(T(v)) = (I_WT)(v).$$

Hence

$$TI_V = I_WT = T.$$

(3) Distributive properties. Pick $T, T_1, T_2 \in \mathcal{L}(U, V)$ and $S, S_1, S_2 \in \mathcal{L}(V, W)$. Then for any $u \in U$,

$$\begin{aligned}((S_1 + S_2)T)(u) &= S_1(T(u)) + S_2(T(u)) \\ &= (S_1T)(u) + (S_2T)(u) \\ &= (S_1T + S_2T)(u)\end{aligned}$$

and

$$\begin{aligned}(S(T_1 + T_2))(u) &= S(T_1(u) + T_2(u)) \\ &= S(T_1(u)) + S(T_2(u)) \\ &= (ST_1)(u) + (ST_2)(u) \\ &= (ST_1 + ST_2)(u).\end{aligned}$$

Hence

$$(S_1 + S_2)T = S_1T + S_2T$$

and

$$S(T_1 + T_2) = ST_1 + ST_2.$$

□

7.4. Exercise 3.A.7: Linear Map on 1D Space is Scalar Multiplication

Show that every linear map from a 1-dimensional vector space to itself is multiplication by some scalar. More precisely, prove that if $\dim V = 1$ and $T \in \mathcal{L}(V, V)$, then there exists $\lambda \in \mathbb{F}$ such that $Tv = \lambda v$ for all $v \in V$.

Proof. Since $\dim V = 1$, there exists a vector b which is a basis of V. Pick any $T \in \mathcal{L}(V, V)$, then, by the definition of basis, there exists $\lambda \in \mathbb{F}$ such that

$$T(b) = \lambda b.$$

We claim that $Tv = \lambda v$ for all $v \in V$. To show this, pick any $v \in V$, then there exists $\alpha \in \mathbb{F}$ such that

$$v = \alpha b.$$

Hence

$$T(v) = T(\alpha b) = \alpha T(b) = \alpha\lambda b = \lambda\alpha b = \lambda v.$$

□

7.5. Exercise 3.A.8: Homogeneity Alone does NOT Imply Linearity

Give an example of a function $\varphi : \mathbb{R}^2 \to \mathbb{R}$ such that

$$\varphi(av) = a\varphi(v)$$

for all $a \in \mathbb{R}$ and all $v \in \mathbb{R}^2$ but φ is not linear.

Proof. Let $\varphi(x, y) \equiv (x^3 + y^3)^{\frac{1}{3}}$ where $(x, y) \in \mathbb{R}^2$.

(1) Pick any $a \in \mathbb{R}$, then

$$\begin{aligned}\varphi(ax, ay) &= (a^3x^3 + a^3y^3)^{\frac{1}{3}} \\ &= (a^3(x^3 + y^3))^{\frac{1}{3}} \\ &= a(x^3 + y^3)^{\frac{1}{3}} \\ &= a\varphi(x, y).\end{aligned}$$

(2) φ is not linear because

$$\begin{aligned}\varphi((1,2)+(3,4)) &= (4^3+6^3)^{\frac{1}{3}} \\ &= 280^{\frac{1}{3}} \\ &\neq 9^{\frac{1}{3}} + 91^{\frac{1}{3}} \\ &= \varphi(1,2)+\varphi(3,4)\end{aligned}$$

□

7.6. Exercise 3.A.9: Additivity Alone does NOT Imply Linearity

Give an example of a function $\varphi : \mathbb{C} \to \mathbb{C}$ such that

$$\varphi(w+z) = \varphi(w) + \varphi(z)$$

for all $w, z \in \mathbb{C}$ but φ is not linear.

Proof. Let $\varphi(z) = \Re(z)$ where $z \in \mathbb{C}$.

(1) Pick any $w, z \in \mathbb{C}$, then

$$\begin{aligned}\varphi(w+z) &= \Re(w+z) \\ &= \Re(w) + \Re(z) \\ &= \varphi(w) + \varphi(z).\end{aligned}$$

(2) φ is not linear because

$$\begin{aligned}i\varphi(i) &= i\Re(i) \\ &= 0 \\ &\neq -1 \\ &= \Re(-1) \\ &= \varphi(i \cdot i).\end{aligned}$$

□

CHAPTER 8

Null Spaces and Ranges

8.1. Exercise 3.B.1: A Linear Map T with $\dim \mathbf{null}\ T = 3, \dim \mathbf{range}\ T = 2$

Give an example of a linear map T such that

$$\dim \operatorname{null} T = 3 \text{ and } \dim \operatorname{range} T = 2.$$

Proof. Define T as the linear map in $\mathcal{L}(\mathbb{R}^5, \mathbb{R}^5)$ such that

$$T(a, b, c, d, e) = (a, b, 0, 0, 0)$$

for all $(a, b, c, d, e) \in \mathbb{R}^5$.

It is clear that range $T = 2$ and $\dim V = 5$. Hence, by the fundamental theorem of linear maps,

$$\dim \operatorname{null} T = \dim V - \dim \operatorname{range} T = 3.$$

□

8.2. Exercise 3.B.2: range $S \subset$ null T Implies $(ST)^2 = 0$

Suppose V is a vector space and $S, T \in \mathcal{L}(V, V)$ are such that

$$\operatorname{range} S \subset \operatorname{null} T.$$

Prove that $(ST)^2 = 0$.

Proof. Pick any $v \in V$, then

$$\begin{gathered} v \in V \\ \Downarrow \\ T(v) \in V \\ \Downarrow \\ S(T(v)) \in \operatorname{range} S \\ \Downarrow \\ S(T(v)) \in \operatorname{null} T \\ \Downarrow \\ T(S(T(v))) = 0 \\ \Downarrow \\ S(T(S(T(v)))) = 0. \end{gathered}$$

Hence $(ST)^2 = 0$. □

8.3. Exercise 3.B.3: Spanning = Surjective; Independent = Injective

Suppose $v_1, \ldots, v_m$ is a list of vectors in V. Define $T \in \mathcal{L}(\mathbb{F}^m, V)$ by

$$T(z_1, \ldots, z_m) = z_1 v_1 + \cdots + z_m v_m.$$

(a) What property of T corresponds to $v_1, \ldots, v_m$ spanning V?
(b) What property of T corresponds to $v_1, \ldots, v_m$ being linearly independent?

PROOF. (a) We claim that T being surjective corresponds to $v_1, \ldots, v_m$ spanning V.

If T is surjective, then range $T = V$. Since range $T = \operatorname{span}(v_1, \ldots, v_m)$, we conclude that $v_1, \ldots, v_m$ spans V.

If $v_1, \ldots, v_m$ spans V, then range $T = V$. Hence T is surjective.

(b) We claim that T being injective corresponds to $v_1, \ldots, v_m$ being linearly independent.

If T is injective, then 3.16 of [**Axl14**] implies that null $T = \{0\}$. Hence $T(z_1, \ldots, z_m) = z_1 v_1 + \cdots + z_m v_m = 0$ implies $z_1 = \cdots = z_m = 0$. Therefore $v_1, \ldots, v_m$ is linearly independent.

If $v_1, \ldots, v_m$ is linearly independent, then $z_1 v_1 + \cdots + z_m v_m = 0$ implies $z_1 = \cdots = z_m = 0$. Hence $T(z_1, \ldots, z_m) = 0$ implies $z_1 = \cdots = z_m = 0$. By 3.16 of [**Axl14**], T is therefore injective.

□

8.4. Exercise 3.B.4: $\{T \in \mathcal{L}(\mathbb{R}^5, \mathbb{R}^4) : \dim \textbf{null } T > 2\}$ is NOT a Subspace

Show that

$$\{T \in \mathcal{L}(\mathbb{R}^5, \mathbb{R}^4) : \dim \operatorname{null} T > 2\}$$

is not a subspace of $\mathcal{L}(\mathbb{R}^5, \mathbb{R}^4)$.

PROOF. Let S, T be the linear maps in $\mathcal{L}(\mathbb{R}^5, \mathbb{R}^4)$ such that

$$S(x_1, x_2, x_3, x_4, x_5) = (x_1, x_2, 0, 0)$$

and

$$T(x_1, x_2, x_3, x_4, x_5) = (x_1, 0, x_3, 0)$$

for all $(x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5$. Then

$$(S + T)(x_1, x_2, x_3, x_4, x_5) = (2x_1, x_2, x_3, 0)$$

for all $(x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5$.

Clearly, $\dim \operatorname{range} S = \dim \operatorname{range} T = 2$ and $\dim \operatorname{range} (S + T) = 3$. Hence the fundamental theorem of linear maps implies

$$\dim \operatorname{null} S = \dim \operatorname{null} T = 5 - 2 = 3 > 2$$

and

$$\dim \operatorname{null} (S + T) = 5 - 3 = 2 \not> 2.$$

We have thus shown that

$$\{T \in \mathcal{L}(\mathbb{R}^5, \mathbb{R}^4) : \dim \operatorname{null} T > 2\}$$

is not closed under addition, hence it is not a subspace of $\mathcal{L}(\mathbb{R}^5, \mathbb{R}^4)$. □

8.5. Exercise 3.B.5: A Linear Map T such that range $T =$ null T

Give an example of a linear map $T : \mathbb{R}^4 \to \mathbb{R}^4$ such that

$$\operatorname{range} T = \operatorname{null} T.$$

PROOF. Let T be the linear map in $\mathcal{L}(\mathbb{R}^4, \mathbb{R}^4)$ such that

$$T(x_1, x_2, x_3, x_4) = (0, 0, x_1, x_2)$$

for all $(x_1, x_2, x_3, x_4) \in \mathbb{R}^4$. Then

$$\operatorname{range} T = \{(0, 0, x_1, x_2) : x_1, x_2 \in \mathbb{R}\}$$

and

$$\operatorname{null} T = \{(0, 0, x_3, x_4) : x_3, x_4 \in \mathbb{R}\}.$$

Hence

$$\text{range } T = \text{null } T.$$

$\square$

CHAPTER 9

Matrices

9.1. Exercise 3.C.1: $\mathcal{M}(T)$ Has At Least $\dim \operatorname{range} T$ Nonzero Entries

Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Show that with respect to each choice of bases of V and W, the matrix of T has at least $\dim \operatorname{range} T$ nonzero entries.

Proof. Choose any basis $v_1, \ldots, v_n$ of V and basis $w_1, \ldots, w_m$ of W, suppose the matrix of T with respect to those bases has k nonzero entries where

$$k < \dim \operatorname{range} T. \tag{9.1}$$

Then the matrix has at most k nonzero columns. Consequently, by the definition of matrix of a linear map, at most k vectors of $Tv_1, \ldots, Tv_n$ are nonzero. Hence

$$\dim \operatorname{range} T \leqslant k,$$

which contradicts with (9.1). □

9.2. Exercise 3.C.2: Find Bases from the Matrix of $D \in \mathcal{L}(\mathcal{P}_3(\mathbb{R}), \mathcal{P}_2(\mathbb{R}))$

Suppose $D \in \mathcal{L}(\mathcal{P}_3(\mathbb{R}), \mathcal{P}_2(\mathbb{R}))$ is the differentiation map defined by $Dp = p'$. Find a basis of $\mathcal{P}_3(\mathbb{R})$ and a basis of $\mathcal{P}_2(\mathbb{R}))$ such that the matrix of D with respect to these bases is

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

Proof. Define

$$w_1 \equiv 1, w_2 \equiv x, w_3 \equiv x^2$$

and

$$v_1 \equiv x, v_2 \equiv \frac{1}{2}x^2, v_3 \equiv \frac{1}{3}x^3, v_4 \equiv 1.$$

Clearly, w_1, w_2, w_3 is a basis of $\mathcal{P}_2(\mathbb{R})$ and v_1, v_2, v_3, v_4 is a basis of $\mathcal{P}_3(\mathbb{R})$. Moreover,

$$Dv_1 = w_1, Dv_2 = w_2, Dv_3 = w_3, Dv_4 = 0.$$

Therefore the matrix of D under these bases is

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

□

9.3. Exercise 3.C.3: Find Bases for V, W from $\mathcal{M}(T)$ where $T \in \mathcal{L}(V, W)$

Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that there exist a basis of V and a basis of W such that with respect to these bases, all entries of $\mathcal{M}(T)$ are 0 except that the entries in row j, column j, equal 1 for $1 \leqslant j \leqslant \dim \operatorname{range} T$.

Proof. Let $u_1, \ldots, u_m$ be a basis of null T; thus $\dim \operatorname{null} T = m$. By 2.33 of [**Axl14**], the linearly independent list $u_1, \ldots, u_m$ can be extended to a basis

$$v_1, \ldots, v_n, u_1, \ldots, u_m \tag{9.2}$$

of V. Thus $\dim V = m + n$. It is easy to verify that $Tv_1, \ldots, Tv_n$ is a basis of range T. Again, by 2.33 of [**Axl14**], the linearly independent list $Tv_1, \ldots, Tv_n$ can be extended to a basis

$$Tv_1, \ldots, Tv_n, s_1, \ldots, s_l \tag{9.3}$$

of W. Thus $\dim W = n + l$. The matrix of T under bases (9.2) and (9.3) is then of size $(n + l) \times (n + m)$, and all of its entries equal to 0 except that the entries in row j, column j, equal 1 for $1 \leqslant j \leqslant n$. □

9.4. Exercise 3.C.4: Find Bases for W from $\mathcal{M}(T)$ where $T \in \mathcal{L}(V, W)$

Suppose $v_1, \ldots, v_m$ is a basis of V and W is finite-dimensional. Suppose $T \in \mathcal{L}(V, W)$. Prove that there exists a basis $w_1, \ldots, w_n$ of W such that all the entries in the first column of $\mathcal{M}(T)$ (with respect to the bases $v_1, \ldots, v_m$ and $w_1, \ldots, w_n$) are 0 except for possibly a 1 in the first row, first column.

Proof. If $Tv_1 = 0$, then any basis of W would make all the entries in the first column of $\mathcal{M}(T)$ equal to 0.

If $Tv_1 \neq 0$, then let $w_1 \equiv Tv_1$. By 2.33 of [**Axl14**], the linearly independent list w_1 can be extended to a basis $w_1, \ldots, w_n$ of W. Under this basis, all the entries in the first column of $\mathcal{M}(T)$ are 0 except for a 1 in the first row, first column. □

9.5. Exercise 3.C.5: Find Bases for V from $\mathcal{M}(T)$ where $T \in \mathcal{L}(V, W)$

Suppose $w_1, \ldots, w_n$ is a basis of W and V is finite-dimensional. Suppose $T \in \mathcal{L}(V, W)$. Prove that there exists a basis $v_1, \ldots, v_m$ of V such that all the entries in the first row of $\mathcal{M}(T)$ (with respect to the bases $v_1, \ldots, v_m$ and $w_1, \ldots, w_n$) are 0 except for possibly a 1 in the first row, first column.

Proof. Pick any basis

$$v_1, \ldots, v_m$$

of V. Denote M to be the matrix of T under this basis. If all the entries in the first row of M are 0, then the proof is done.

Otherwise, suppose column j of the first row of M contains a nonzero entry. Let

$$v_1' \equiv \frac{1}{M_{1,j}} v_j$$

and

$$\{v_2', \ldots, v_m'\} \equiv \{v_1, \ldots, v_{j-1}, v_{j+1}, \ldots, v_m\}.$$

Then $v_1', \ldots, v_m'$ is also a basis of V.

Denote M' as the matrix of T under the basis

$$v_1', \ldots, v_m'.$$

Further, let

$$\begin{aligned} v_1'' &\equiv v_1', \\ v_2'' &\equiv v_2' - M_{1,2}' v_1', \\ &\vdots \\ v_m'' &\equiv v_m' - M_{1,m}' v_1'. \end{aligned}$$

Then $v_1'', \ldots, v_m''$ is also a basis of V. Moreover,

$$\begin{aligned} T(v_1'') = T(v_1') &= T(\frac{1}{M_{1,j}} v_j) \\ &= \frac{1}{M_{1,j}}(M_{1,j} w_1 + M_{2,j} w_2 + \cdots + M_{n,j} w_n) \\ &= w_1 + \frac{M_{2,j}}{M_{1,j}} w_2 + \cdots + \frac{M_{n,j}}{M_{1,j}} w_n. \end{aligned}$$

And for any $i = 2, \ldots, m$,

$$\begin{aligned} T(v_i'') &= T(v_i' - M_{1,i}' v_1') \\ &= T(v_i') - M_{1,i}' T(v_1') \\ &= M_{1,i}' w_1 + \cdots + M_{n,i}' w_n - M_{1,i}'(w_1 + \frac{M_{2,j}}{M_{1,j}} w_2 + \cdots + \frac{M_{n,j}}{M_{1,j}} w_n) \\ &= (M_{2,i}' - \frac{M_{1,i}' M_{2,j}}{M_{1,j}}) w_2 + \cdots + (M_{n,i}' - \frac{M_{1,i}' M_{n,j}}{M_{1,j}}) w_n. \end{aligned}$$

Hence all the entries in the first row of $\mathcal{M}(T)$ with respect to the bases $v_1'', \ldots, v_m''$ and $w_1, \ldots, w_n$ are 0 except for a 1 in the first row, first column. $\square$

CHAPTER 10

Invertibility and Isomorphic Vector Spaces

10.1. Exercise 3.D.1: $(ST)^{-1} = T^{-1}S^{-1}$

Suppose $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$ are both invertible linear maps. Prove that $ST \in \mathcal{L}(U, W)$ is invertible and that $(ST)^{-1} = T^{-1}S^{-1}$.

Proof. Since T and S are invertible, 3.56 of [**Axl14**] implies that T and S are injective and surjective.

Suppose $S(T(u_1)) = S(T(u_2))$ where $u_1, u_2 \in U$, then that S is injective implies $T(u_1) = T(u_2)$, and that T is injective implies $u_1 = u_2$. Hence ST is injective.

Pick any $w \in W$. That S is surjective implies the existence of a $v \in V$ such that $S(v) = w$. Further, that T is surjective implies the existence of a $u \in U$ such that $T(u) = v$. Hence $S(T(u)) = w$. Thus ST is surjective.

Thus 3.56 of [**Axl14**] implies that ST is invertible.

Finally, since

$$(T^{-1}S^{-1})(ST) = T^{-1}(S^{-1}S)T = T^{-1}T = I$$

and

$$(ST)(T^{-1}S^{-1}) = S(TT^{-1})S^{-1} = SS^{-1} = I,$$

we conclude that

$$(ST)^{-1} = T^{-1}S^{-1}.$$

□

10.2. Exercise 3.D.2: Noninvertible Operators on V is NOT a Subspace

Suppose V is finite-dimensional and $\dim V > 1$. Prove that the set of noninvertible operators on V is not a subspace of $\mathcal{L}(V)$.

Proof. Let $v_1, v_2, \ldots, v_n$ be a basis of V where $n \geqslant 2$. Let S be the operator on V defined as

$$S(a_1v_1 + a_2v_2 + \cdots + a_nv_n) = a_1v_1,$$

and let T be the operator on V defined as

$$T(a_1v_1 + a_2v_2 + \cdots + a_nv_n) = a_2v_2 + a_3v_3 + \cdots + a_nv_n.$$

Then S is noninvertible because $v_2 \in \operatorname{null} S$, and T is noninvertible because $v_1 \in \operatorname{null} T$. However, $S + T$ is invertible because $S + T$ equals the identity operator.

We have shown that the set of noninvertible operators on V is not closed under addition, hence it is not a subspace of $\mathcal{L}(V)$. □

10.3. Exercise 3.D.3: Injective Subspace Mappings and Invertible Operators

Suppose V is finite-dimensional, U is a subspace of V, and $S \in \mathcal{L}(U, V)$. Prove there exists an invertible operator $T \in \mathcal{L}(V)$ such that $Tu = Su$ for every $u \in U$ if and only if S is injective.

Proof. Suppose there exists an invertible operator $T \in \mathcal{L}(V)$ such that $Tu = Su$ for every $u \in U$. It follows that

$$T^{-1}(Su) = T^{-1}(Tu) = u \tag{10.1}$$

for every $u \in U$. Pick any $u_1, u_2 \in U$ such that $Su_1 = Su_2$. Then $T^{-1}(Su_1) = T^{-1}(Su_2)$, and (10.1) implies that $u_1 = u_2$. Hence S is injective.

Suppose S is injective. Pick a basis $u_1, \ldots, u_k$ of U and extend it to a basis

$$u_1, \ldots, u_k, v_{k+1}, \ldots, v_n$$

of V; thus $\dim U = k, \dim V = n$. Since S is injective, S is an isomorphism between U and range S. Hence

$$Su_1, \ldots, Su_k$$

is a basis of range T and can be extended to a basis

$$Su_1, \ldots, Su_k, t_{k+1}, \ldots, t_n$$

of V. Define T as the operator on V such that

$$T(u_i) = Su_i \text{ where } i = 1, \ldots, k$$

and

$$T(v_i) = t_i \text{ where } i = k+1, \ldots, n.$$

Clearly, T is invertible and $Tu = Su$ for every $u \in U$. □

10.4. Exercise 3.D.4: null T_1 = null T_2 if and only if $T_1 = ST_2$

Suppose W is finite-dimensional and $T_1, T_2 \in \mathcal{L}(V, W)$. Prove that null T_1 = null T_2 if and only if there exists an invertible operator $S \in \mathcal{L}(W)$ such that $T_1 = ST_2$.

Proof. Suppose there exists an invertible operator $S \in \mathcal{L}(W)$ such that $T_1 = ST_2$, then

$$S^{-1}T_1 = T_2.$$

Pick any $v_1 \in \text{null } T_1$, then $T_1 v_1 = 0$ and

$$T_2 v_1 = S^{-1}T_1(v_1) = 0.$$

Hence null $T_1 \subset$ null T_2. Also, pick any $v_2 \in \text{null } T_2$, then

$$T_1 v_2 = ST_2(v_2) = 0.$$

Hence null $T_2 \subset$ null T_1. Therefore

$$\text{null } T_1 = \text{null } T_2.$$

On the other hand, suppose null T_1 = null T_2. Denote

$$N \equiv \text{null } T_1 = \text{null } T_2$$

and

$$m \equiv \dim W.$$

Pick a basis $w_1, \ldots, w_k$ of range T_2 and extend it to a basis of W:

$$w_1, \ldots, w_k, w_{k+1}, \ldots, w_m.$$

Then there exists nonzero vectors $v_1, \ldots, v_k \in V$ such that

$$T_2 v_i = w_i, \quad \forall i \in \{1, \ldots, k\}.$$

Then it is easy to verify that

$$V = N \oplus \operatorname{span}(v_1, \ldots, v_k).$$

and

$$T_1 v_1, \ldots, T_1 v_k$$

is a linearly independent list, which can be extended to a basis of W:

$$T_1 v_1, \ldots, T_1 v_k, w'_{k+1}, \ldots, w'_m.$$

Let $S \in \mathcal{L}(W)$ be defined by

$$S(w_i) \equiv T_1 v_i, \quad \forall i \in \{1, \ldots, k\},$$

and

$$S(w_i) \equiv w'_i, \quad \forall i \in \{k+1, \ldots, m\}.$$

We claim that $T_1 = ST_2$. To show this, pick any $v \in V$. Then there exists $v_0 \in N$ and $a_1, \ldots, a_k \in \mathbb{F}$ such that

$$v = v_0 + a_1 v_1 + \cdots + a_k v_k$$

and

$$\begin{aligned} T_1(v) &= T_1(v_0 + a_1 v_1 + \cdots + a_k v_k) \\ &= a_1 T_1 v_1 + \cdots + a_k T_1 v_k \\ &= a_1 S w_1 + \cdots + a_k S w_k \\ &= S(a_1 w_1 + \cdots + a_k w_k) \\ &= S(T_2(a_1 v_1 + \cdots + a_k v_k)) \\ &= S(T_2(v_0 + a_1 v_1 + \cdots + a_k v_k)) \\ &= ST_2(v). \end{aligned}$$

□

10.5. Exercise 3.D.5: range T_1 = range T_2 if and only if $T_1 = T_2 S$

Suppose V is finite-dimensional and $T_1, T_2 \in \mathcal{L}(V, W)$. Prove that range T_1 = range T_2 if and only if there exists an invertible operator $S \in \mathcal{L}(V)$ such that $T_1 = T_2 S$.

Proof. Suppose there exists an invertible operator $S \in \mathcal{L}(V)$ such that $T_1 = T_2 S$. Pick any $w \in$ range T_1, then there exists $v_1 \in V$ such that $T_1 v_1 = w$. Hence $T_2 S v_1 = w$ and $w \in$ range T_2. Also, pick any $w \in$ range T_2, then there exists $v_2 \in V$ such that $T_2 v_2 = w$. Hence $T_1 S^{-1} v_2 = w$ and $w \in$ range T_1. Therefore range T_1 = range T_2.

On the other hand, suppose range T_1 = range T_2. Denote

$$R \equiv \text{range } T_1 = \text{range } T_2$$

and

$$n \equiv \dim V.$$

We claim that R is finite-dimensional. To show this, pick a basis $u_1, \ldots, u_k$ of null T_1 and extend it to a basis of V:

$$u_1, \ldots, u_k, u_{k+1}, \ldots, u_n.$$

Then it is easy to verify that

$$T_1 u_{k+1}, \ldots, T_1 u_n$$

is a basis of R; thus

$$\dim R = n - k.$$

The fundamental theorem of linear maps then implies

$$k = \dim \operatorname{null} T_1 = \dim V - \dim R = \dim \operatorname{null} T_2.$$

Moreover, there exist $v_{k+1}, \ldots, v_n \in V$ such that

$$T_2 v_i = T_1 u_i, \quad \forall i \in \{k+1, \ldots, n\}.$$

Pick a basis $v_1, \ldots, v_k$ of null T_2. We claim that

$$v_1, \ldots, v_k, v_{k+1}, \ldots, v_n$$

is a basis of V. By 2.39 of [**Axl14**], it suffices to show its linear independence. Suppose there exists $a_1, \ldots, a_n \in \mathbb{F}$ such that

$$a_1 v_1 + \cdots + a_n v_n = 0.$$

Then

$$\begin{gathered}
T_2(a_1 v_1 + \cdots + a_n v_n) = 0 \\
\Downarrow \\
a_{k+1} T_2 v_{k+1} + \cdots + a_n T_2 v_n = 0 \\
\Downarrow \\
a_{k+1} T_1 u_{k+1} + \cdots + a_n T_1 u_n = 0 \\
\Downarrow \\
a_{k+1} = \cdots = a_n = 0 \\
\Downarrow \\
a_1 v_1 + \cdots + a_k v_k = 0 \\
\Downarrow \\
a_1 = \cdots = a_k = 0.
\end{gathered}$$

Let $S \in \mathcal{L}(V)$ be defined by

$$S(u_i) \equiv v_i, \quad \forall i \in \{1, \ldots, n\}.$$

We claim that $T_1 = T_2 S$. To show this, pick any $v \in V$. Then there exists $a_1, \ldots, a_n \in \mathbb{F}$ such that

$$v = a_1 u_1 + \cdots + a_n u_n.$$

Then

$$\begin{aligned}
T_1(v) &= T_1(a_1 u_1 + \cdots + a_n u_n) \\
&= a_{k+1} T_1 u_{k+1} + \cdots + a_n T_1 u_n \\
&= a_{k+1} T_2 v_{k+1} + \cdots + a_n T_2 v_n \\
&= T_2(a_{k+1} v_{k+1} + \cdots + a_n v_n) \\
&= T_2(a_{k+1} S u_{k+1} + \cdots + a_n S u_n) \\
&= T_2(S(a_{k+1} u_{k+1} + \cdots + a_n u_n)) \\
&= T_2(S(v)).
\end{aligned}$$

□

CHAPTER 11

Products and Quotients of Vector Spaces

11.1. Exercise 3.E.1: T is a Linear Map iff graph of T is a Subspace of $V \times W$

Suppose T is a function from V to W. The graph of T is the subset of $V \times W$ defined by

$$\text{graph of } T = \{(v, Tv) \in V \times W : v \in V\}.$$

Prove that T is a linear map if and only if the graph of T is a subspace of $V \times W$.

Proof. Suppose T is a linear map. By 3.11 of [**Axl14**], $T(0) = 0$. Hence

$$(0, 0) \in \text{graph of } T.$$

Suppose $(u, Tu), (w, Tw) \in$ graph of T. Then $T(u + w) = Tu + Tw \in W$ implies

$$(u + w, Tu + Tw) \in \text{graph of } T.$$

Suppose $a \in \mathbb{F}$ and $(u, Tu) \in$ graph of T. Then $T(au) = aTu$ implies

$$(au, aTu) \in \text{graph of } T.$$

By 13.4 of [**Axl14**], we therefore conclude that the graph of T is a subspace of $V \times W$.

On the other hand, suppose the graph of T is a subspace of $V \times W$. Pick any $u, v \in V$ such that $(u, Tu), (v, Tv) \in$ graph of T. Then $(u + v, Tu + Tv) \in$ graph of T. Hence

$$T(u + v) = Tu + Tv.$$

Pick any $a \in \mathbb{F}$, then $(au, aTu) \in$ graph of T. Hence

$$T(au) = aTu.$$

Therefore, by definition, we conclude that T is a linear map. □

11.2. Exercise 3.E.2: V_j is Finite-dimensional if $V_1 \times \cdots \times V_m$ is Finite-dimensional

Suppose $V_1, \ldots, V_m$ are vector spaces such that $V_1 \times \cdots \times V_m$ is finite-dimensional. Prove that V_j is finite-dimensional for each $j = 1, \ldots, m$.

Proof. Suppose there exists $j \in \{1, \ldots, m\}$ such that V_j is infinite-dimensional. Then, by 4.14, there is a sequence $v_{j1}, v_{j2}, \ldots$ of vectors in V_j such that $v_{j1}, v_{j2}, \ldots, v_{jk}$ is linearly independent for every positive integer k.

Let the sequence $v_1, v_2, \ldots$ of vectors in $V_1 \times \cdots \times V_m$ be defined by: for each positive integer i, the vector v_i has its jth slot equal to v_{ji} and other slots equal to 0. Then $v_1, v_2, \ldots, v_k$ is linearly independent for every positive integer k. Therefore, again by 4.14, $V_1 \times \cdots \times V_m$ is infinite-dimensional. □

11.3. Exercise 3.E.3: Example of $U_1 \times U_2$ Isomorphic to Non-Direct-Sum $U_1 + U_2$

Give an example of a vector space V and subspaces U_1, U_2 of V such that $U_1 \times U_2$ is isomorphic to $U_1 + U_2$ but $U_1 + U_2$ is not a direct sum.

Proof. Let $V \equiv U_1 \equiv \mathbb{R}^\infty$. Let

$$U_2 \equiv \{(x, 0, 0, \dots) : x \in \mathbb{R}\}.$$

Then V is a vector space and U_1, U_2 are two subspaces of V, and

$$U_1 + U_2 = \mathbb{R}^\infty.$$

Since $U_1 \cap U_2 = U_2 \neq \{0\}$, $U_1 + U_2$ is not a direct sum. Let function $T : U_1 \times U_2 \to U_1 + U_2$ be defined by

$$T((x, 0, 0, \dots), (y_1, y_2, \dots)) = (x, y_1, y_2, \dots).$$

We claim that T is a linear map. Suppose

$$u = ((u_x, 0, 0, \dots), (u_{y_1}, u_{y_2}, \dots))$$

and

$$v = ((v_x, 0, 0, \dots), (v_{y_1}, v_{y_2}, \dots)).$$

Then

$$\begin{aligned} T(u + v) &= (u_x + v_x, u_{y_1} + v_{y_1}, u_{y_2} + v_{y_2}, \dots) \\ &= (u_x, u_{y_1}, u_{y_2}, \dots) + (v_x, v_{y_1}, v_{y_2}, \dots) \\ &= T(u) + T(v). \end{aligned}$$

Pick any $a \in \mathbb{F}$,

$$\begin{aligned} T(au) &= (au_x, au_{y_1}, au_{y_2}, \dots) \\ &= a(u_x, u_{y_1}, u_{y_2}, \dots) \\ &= aT(u). \end{aligned}$$

Clearly T is both injective and surjective. Therefore, $U_1 \times U_2$ is isomorphic to $U_1 + U_2$. □

11.4. Exercise 3.E.4: $\mathcal{L}(V_1 \times \cdots \times V_m, W)$ and $\mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W)$ are Isomorphic

Suppose $V_1, \dots, V_m$ are vector spaces. Prove that $\mathcal{L}(V_1 \times \cdots \times V_m, W)$ and $\mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W)$ are isomorphic vector spaces.

Proof. Let function $T : \mathcal{L}(V_1 \times \cdots \times V_m, W) \to \mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W)$ be defined by

$$T(P) = (P_1, \dots, P_m)$$

where

$$\begin{aligned} P_1(v_1) &= P(v_1, 0, \dots, 0) \quad \forall v_1 \in V_1, \\ P_2(v_2) &= P(0, v_2, \dots, 0) \quad \forall v_2 \in V_2, \\ &\vdots \\ P_m(v_m) &= P(0, 0, \dots, v_m) \quad \forall v_m \in V_m. \end{aligned}$$

We claim that T is a linear map. Pick any $P_1, P_2 \in \mathcal{L}(V_1 \times \cdots \times V_m, W)$, then

$$\begin{aligned} T(P_1 + P_2) &= ((P_1 + P_2)_1, \dots, (P_1 + P_2)_m) \\ &= ((P_1)_1 + (P_2)_1, \dots, (P_1)_m + (P_2)_m) \\ &= ((P_1)_1, \dots, (P_1)_m) + ((P_2)_1, \dots, (P_2)_m) \\ &= T(P_1) + T(P_2). \end{aligned}$$

Pick any $\lambda \in \mathbb{F}, P \in \mathcal{L}(V_1 \times \cdots \times V_m, W)$, then

$$\begin{aligned} T(\lambda P) &= ((\lambda P)_1, \ldots, (\lambda P)_m) \\ &= \lambda(P_1, \ldots, P_m) \\ &= \lambda T(P). \end{aligned}$$

We claim that T is injective. Suppose $T(P) = 0$. Then,

$$\begin{aligned} P(v_1, 0, \ldots, 0) &= 0 \quad \forall v_1 \in V_1, \\ P(0, v_2, \ldots, 0) &= 0 \quad \forall v_2 \in V_2, \\ &\vdots \\ P(0, 0, \ldots, v_m) &= 0 \quad \forall v_m \in V_m. \end{aligned}$$

Hence, for all $v_1 \in V_1, \ldots, v_m \in V_m$,

$$P(v_1, \ldots, v_m) = P(v_1, \ldots, 0) + \cdots + P(0, \ldots, v_m) = 0.$$

Thus

$$P = 0.$$

We claim that T is surjective. Pick any $(P_1, \ldots, P_m) \in \mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W)$, let $P \in \mathcal{L}(V_1 \times \cdots \times V_m, W)$ be defined by

$$P(v_1, \ldots, v_m) = P_1(v_1) + \cdots + P_m(v_m) \quad \forall v_1 \in V_1, \ldots, v_m \in V_m.$$

Then

$$T(P) = (P_1, \ldots, P_m).$$

Therefore T is an isomorphism from $\mathcal{L}(V_1 \times \cdots \times V_m, W)$ to $\mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W)$. □

11.5. Exercise 3.E.5: $\mathcal{L}(V, W_1 \times \cdots \times W_m)$ and $\mathcal{L}(V, W_1) \times \cdots \times \mathcal{L}(V, W_m)$ are Isomorphic

Suppose $W_1, \ldots, W_m$ are vector spaces. Prove that $\mathcal{L}(V, W_1 \times \cdots \times W_m)$ and $\mathcal{L}(V, W_1) \times \cdots \times \mathcal{L}(V, W_m)$ are isomorphic vector spaces.

Proof. Let

$$\begin{aligned} \Pi_1 &\in \mathcal{L}(W_1 \times \cdots \times W_m, W_1), \\ \Pi_2 &\in \mathcal{L}(W_1 \times \cdots \times W_m, W_2), \\ &\vdots \\ \Pi_m &\in \mathcal{L}(W_1 \times \cdots \times W_m, W_m) \end{aligned}$$

be defined by

$$\begin{aligned} \Pi_1(w_1, \ldots, w_m) &= w_1, \\ \Pi_2(w_1, \ldots, w_m) &= w_2, \\ &\vdots \\ \Pi_m(w_1, \ldots, w_m) &= w_m \end{aligned}$$

for all $(w_1, \ldots, w_m) \in W_1 \times \cdots \times W_m$.

Let function $T : \mathcal{L}(V, W_1 \times \cdots \times W_m) \to \mathcal{L}(V, W_1) \times \cdots \times \mathcal{L}(V, W_m)$ be defined by

$$T(P) = (P_{\Pi_1}, \ldots, P_{\Pi_m})$$

where

$$\begin{aligned} P_{\Pi_1}(v) &= \Pi_1(P(v)), \\ P_{\Pi_2}(v) &= \Pi_2(P(v)), \\ &\vdots \\ P_{\Pi_m}(v) &= \Pi_m(P(v)). \end{aligned}$$

for all $v \in V$.

We claim that T is a linear map. Pick any $P_1, P_2 \in \mathcal{L}(V, W_1 \times \cdots \times W_m)$, then

$$\begin{aligned} T(P_1 + P_2) &= ((P_1 + P_2)_{\Pi_1}, \ldots, (P_1 + P_2)_{\Pi_m}) \\ &= ((P_1)_{\Pi_1} + (P_2)_{\Pi_1}, \ldots, (P_1)_{\Pi_m} + (P_2)_{\Pi_m}) \\ &= ((P_1)_{\Pi_1}, \ldots, (P_1)_{\Pi_m}) + ((P_2)_{\Pi_1}, \ldots, (P_2)_{\Pi_m}) \\ &= T(P_1) + T(P_2). \end{aligned}$$

Pick any $\lambda \in \mathbb{F}, P \in \mathcal{L}(V_1 \times \cdots \times V_m, W)$, then

$$\begin{aligned} T(\lambda P) &= ((\lambda P)_{\Pi_1}, \ldots, (\lambda P)_{\Pi_m}) \\ &= \lambda(P_{\Pi_1}, \ldots, P_{\Pi_m}) \\ &= \lambda T(P). \end{aligned}$$

We claim that T is injective. Suppose $T(P) = 0$. Then,

$$\begin{aligned} \Pi_1(P(v)) &= 0, \\ \Pi_2(P(v)) &= 0, \\ &\vdots \\ \Pi_m(P(v)) &= 0. \end{aligned}$$

for all $v \in V$. Hence, $P(v) = 0$ for all $v \in V$. Thus

$$P = 0.$$

We claim that T is surjective. Pick any $(P_1, \ldots, P_m) \in \mathcal{L}(V, W_1) \times \cdots \times \mathcal{L}(V, W_m)$, let $P \in \mathcal{L}(V, W_1 \times \cdots \times W_m)$ be defined by

$$P(v) = (P_1(v), \ldots, P_m(v_m)) \quad \forall v \in V.$$

Then

$$T(P) = (P_1, \ldots, P_m).$$

Therefore T is an isomorphism from $\mathcal{L}(V, W_1 \times \cdots \times W_m)$ to $\mathcal{L}(V, W_1) \times \cdots \times \mathcal{L}(V, W_m)$. $\square$

CHAPTER 12

Duality

12.1. Exercise 3.F.1: Linear Functionals are either Surjective or Zero

Explain why every linear functional is either surjective or the zero map.

PROOF. Suppose φ is a linear functional from V to $\mathbb{F}$ and φ is neither surjective nor the zero map. Then there exists $y_0 \in \mathbb{F}$ such that

$$\varphi(v) \neq y_0, \quad \forall v \in V.$$

Also, there exist $x_0 \in \mathbb{F}, v_0 \in V$ such that

$$x_0 \neq 0 \text{ and } \varphi(v_0) = x_0.$$

Since φ is a linear map,

$$\begin{aligned}\varphi(\frac{y_0}{x_0} \cdot v_0) &= \frac{y_0}{x_0} \cdot \varphi(v_0) \\ &= \frac{y_0}{x_0} \cdot x_0 \\ &= y_0,\end{aligned}$$

which contradicts with

$$\varphi(v) \neq y_0, \quad \forall v \in V.$$

□

12.2. Exercise 3.F.2: Examples of Linear Functionals on $\mathbb{R}^{[0,1]}$

Give three distinct examples of linear functionals on $\mathbb{R}^{[0,1]}$.

PROOF. Pick any $c \in \mathbb{R}$. Let $\varphi : \mathbb{R}^{[0,1]} \to \mathbb{R}$ be defined by

$$\varphi(f) = f(c), \quad \forall f(x) \in \mathbb{R}^{[0,1]}.$$

Then we claim that φ is a linear map. To show this, pick any $g(x), f(x) \in \mathbb{R}^{[0,1]}$. Then

$$\varphi(f+g) = f(c) + g(c) = \varphi(f) + \varphi(g).$$

Also, pick any $\lambda \in \mathbb{R}$. Then

$$\varphi(\lambda f) = (\lambda f)(c) = \lambda(f(c)) = \lambda\varphi(f).$$

Let $c \equiv 1, 2, 3$ respectively, we then obtained three distinct linear functionals on $\mathbb{R}^{[0,1]}$. □

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